

Quits, Layoffs, and their Destinations

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Abstract

We construct monthly Current Population Survey series that classify every separation into non-employment by both its reason (quit or layoff) and its destination (unemployment or non-participation) from 1978 on. Standard destination flows mischaracterize churn: over a third of laid-off workers exit the labor force, leading employment-to-unemployment (EU) flows to understate true layoffs by 20% to mask a stabilizing negative correlation between quits and layoffs. Properly measured, layoffs account for 62% of employment-population volatility and 35% more of unemployment volatility than EU flows, amplified by a new channel: laid-off workers' rising propensity to flow to unemployment during recessions.

JEL Classification: E24, E32, J63, J64

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1 Introduction

The flows of workers between employment, unemployment, and non-participation are fundamental to understanding labor market dynamics and business cycles. Since [Blanchard and Diamond \(1990\)](#), economists have relied on gross worker flows to analyze unemployment fluctuations, with [Shimer \(2012\)](#) concluding that job finding rates dominate separation rates in driving cyclical unemployment variation. However, this literature has been constrained by a critical limitation: the inability to distinguish voluntary quits from involuntary layoffs in large timely surveys, leading to the conventional assumption that flows from employment to unemployment represent layoffs while flows to non-participation represent quits.

This paper constructs a comprehensive monthly time series that tracks both the reason for a job separation into non-employment (quit versus layoff) and the worker's subsequent participation decision (unemployment versus non-participation), using Current Population Survey (CPS)

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data from 1978 to 2024. Our construction follows [Flaim \(1973\)](#) and creates monthly, seasonally-adjusted series of quits and layoffs closely following the methodology used to produce the Integrated Public Use Microdata Series (IPUMS). The series is available in near real time, is updated monthly, and covers nearly five decades and six recessions. By contrast, the establishment-based quit and layoff series from JOLTS begin only in 2001, and reason-based measures constructed from the SIPP only in the mid-1990s; neither observes a pre-2000 recession. Our contribution is further distinct as we classify the full two-by-two of reason-cross-destination of flows and we released the first data product containing these series that we maintain monthly as a public good.

Our findings challenge conventional wisdom about worker flows and reveal new insights about business cycle dynamics. We find that standard measures of flows between labor market states significantly mischaracterize the composition of job separations. Layoffs to non-employment occur 20% more frequently than employment-to-unemployment (EU) flows. The reason is that over one-third of laid-off workers exit to non-participation rather than active job search, which implies that these workers would be counted in the employment-to-non-participation (EN) flows instead of EU. In the other direction, EN flows overstate quits by 82%, because nearly 40% of classified EN transitions are in fact involuntary.

The distinction between quits and layoffs carries cyclical information that the destination-based flows do not. Although EU and EN flows are essentially uncorrelated, we document a strong negative correlation (-0.46) between quits and layoffs that creates a stabilizing force in the labor market. When layoffs surge during recessions, quits fall sharply, causing total separations to be less volatile than layoffs on their own. This offsetting relationship means that focusing either on total separation rates or on EU alone would understate the underlying volatility in labor demand conditions captured by layoffs.

Using the [Shimer \(2012\)](#) decomposition, we find that properly measured layoffs to non-employment contribute 35% more to unemployment fluctuations than EU flows in a standard decomposition. Further, rises in layoffs drive increases in unemployment at the start of most recessions, a few months before job-finding (UE) rates contribute, which makes tracking layoffs particularly useful for identifying the onset of a recession. This exercise is closest to [Simmons \(2023\)](#), who splits the employment-to-unemployment transition by reason in the SIPP and likewise finds that separations contribute more to unemployment fluctuations once reason is accounted for. We differ in splitting the EU flow itself and in adding the participation destination, the split between unemployment and non-participation, that a reason-based decomposition of EU does not take up. This split reveals a key mechanism that amplifies the role of layoffs in the cyclical volatility of unemployment: laid-off workers are more likely to remain in the labor force during recessions rather than exiting to non-participation. Holding constant the propensity of laid-off workers to remain in the labor force would reduce unemployment volatility by about 20%, with large variance across recessions: changes in participation decisions reduced the increase in unemployment by a third in the 1991 recession but by only 1.5% in the 2020 COVID

spike. This countercyclical attachment of laid-off workers to the labor force is an important but relatively unstudied factor in the volatility of unemployment.

A second and more striking finding concerns employment rather than unemployment. Layoffs account for 62% of the fluctuations in the employment-population ratio, against 30% for the EU flow in Shimer’s own decomposition and 43% for EU in ours. The gap exists because layoffs into non-participation move the employment-population ratio while leaving EU untouched. Over the full 1978 to 2024 sample layoffs also exceed the contribution of job finding from unemployment (50%). The robust message is not that job finding stops mattering, but that the EU flow is a poor proxy for layoffs when employment, rather than unemployment, is the object of interest. This is a subtle but important distinction for policy makers, who should treat layoffs as a first-order factor in the cyclical of employment.

Our decomposition also highlights how offsetting movements in quits dampen the volatility of both unemployment and employment, in fundamentally different ways than the employment-to-non-participation (EN) flow would suggest. Although EN flows are nearly uncorrelated with the unemployment rate, quits are notably procyclical and negatively correlated with it. The procyclicality of quits is large enough to reduce unemployment fluctuations by 60% more than the relatively acyclical EN flow, because many EN flows in fact contain layoffs.

Importantly, we show that our findings are robust to commonly known issues in the Current Population Survey (CPS), such as rotation group bias, misclassification error, and non-response. We identify the issues and show that through appropriate reweighting both the level and cyclical result remain unchanged. In addition, we validate our data using the PSID and SIPP, which observe the reason for separation without imputation, and show that our findings in the CPS are remarkably similar to the other household surveys: the compositional shares agree to within two percentage points, and both surveys independently reproduce the countercyclicality of the layoff-to-non-participation share.

Our empirical patterns hold remarkably consistently across all six NBER recessions in our sample, including the 2020 pandemic recession. Although the causes and magnitudes of recessions differ, the fundamental relationships between quits, layoffs, and their destinations follow the same historical patterns when conditioning on economic fundamentals such as labor market tightness.

The flow rates we measure are equilibrium objects, but the contribution of this paper is to provide an in-depth exposition, validation, and statistical description of the quit-layoff-destination CPS data series we have made publicly available. In a companion paper, [Ellieroth and Michaud \(2026\)](#), we take a fully structural approach in which the destination of a separation arises in equilibrium as a measured signal of a worker’s distance to the participation margin, a close link to labor supply elasticity. We think the statistics in this paper will be useful both for macroeconomists who need a calibration target that is a better analogy of a quit or layoff in their model, and for those trying to better understand labor demand, supply, and markets in

general. The data are freely available online and are updated monthly.¹

It is worth stating plainly what in this paper is new and what corroborates existing work. Three things are new. First, we construct and publicly release the full two-by-two of separations into non-employment by reason (quit or layoff) crossed with destination (unemployment or non-participation), maintained monthly and spanning 1978–2024 — nearly five decades and six recessions, where the establishment-based JOLTS and the SIPP-based reason measures begin only around 2001 and the mid-1990s and observe no pre-2000 recession. Second, we use this cross-classification in a Shimer (2012) decomposition that splits the EU flow itself by reason and adds the participation destination; and in a participation-neutral counterfactual that isolates how the cyclical destination of laid-off workers amplifies unemployment volatility. Third, the four-decade span lets us document the stability of these relationships across all six recessions and a secular decline in separations per unit of tightness that shorter series cannot detect. Two of our findings corroborate and extend prior work. Our measured countercyclicality in the share of layoffs flowing to non-participation is a higher-frequency, reason-based counterpart to the cyclical shift in the wage composition of the unemployment and separation pools documented by Elsby et al. (2015a) and Mueller (2017). Our closest antecedents on the decomposition are Flaim (1969), Flaim (1973), Simmons (2023), and Graves et al. (2025). Flaim (1969) and Flaim (1973) deserve credit as the first to use the same question asked of CPS non-participants to determine whether they quit or were laid-off from their last job that both we and Graves et al. (2025) do. He documents the incidence and cyclicity of labor force exits depending on whether the separation was a quit or layoff. Simmons (2023) splits the EU transition by reason in the SIPP and finds a larger role for separations, which we reproduce and extend by splitting EU within the CPS and adding the participation margin that a reason-only split of EU leaves untouched. Graves et al. (2025) concurrently and independently decompose EN flows into quits and layoffs in the CPS using a methodology that they develop on their own, and study monetary-policy transmission and focus on quits. We differ in that we use established methodology that replicates IPUMS and add the destination margin and build the long, validated, public series. We do not adjudicate the structural mechanism behind cyclical attachment and instead provide the well-measured moments on which such analysis can build.

The remainder of the paper proceeds as follows. Section 2 reviews related literature. Section 3 describes the data and construction of the series, and provides robustness and validation. Section 4 presents the key statistics and their cyclical properties. Section 5 establishes the economic significance of the destination margin, through a participation-neutral counterfactual and a Shimer decomposition. Section 6 compares the six recessions in the sample. Section 7 compares the CPS quits and layoffs to the employer survey (JOLTS), a question we are often asked. Section 8 concludes.

¹Data can be accessed through our website sites.google.com/qlmonthly.com/home, or from the Federal Reserve Bank of St. Louis' FRED as series 738: fred.stlouisfed.org/release?rid=738.

2 Literature Review

Beginning with [Abowd and Zellner \(1985a\)](#) and [Blanchard and Diamond \(1990\)](#), researchers have used the flow approach to understand the evolution of labor market flows across time, cross-sections, and business cycles, and [Rogerson and Shimer \(2011\)](#) summarize how this approach has advanced macroeconomic theory.² A large body of work studies heterogeneity in these flows across worker types and over the cycle, including [Fujita et al. \(Forthcoming\)](#) on direct job-to-job changes, [Ferraro \(2018\)](#) on the skewness of flows, and [Barnichon and Figura \(2015\)](#), [Coglianese \(2018\)](#), [Hall and Kudlyak \(2020\)](#), and [Ellieroth \(2023\)](#). Relative to this work, which computes flows across the three labor-market states, we add a within-flow classification by reason of separation on both the unemployment and the non-participation margins, so that the full reason-by-destination distribution is observed.

Few papers document facts on job separations over the cycle separately for quits and layoffs. [Flaim \(1973\)](#) and [Akerlof et al. \(1988\)](#) established that quits tend to be procyclical and layoffs countercyclical, and we use a quit/layoff classification for the CPS similar to that in [Flaim \(1973\)](#).³ The distinction between classifying separations by their reason and by their destination is due to [Nagypál \(2008\)](#). Most directly related to our measurement, [Graves et al. \(2025\)](#) concurrently and independently decompose employment-to-non-participation flows into quits and layoffs from the CPS. Their study focuses on the relationship between quits and monetary policy shocks. Our measurement adds the destination margin and we produced the first long, validated, publicly available series of these data.

We contribute, second, to the literature that decomposes the contribution of job separation and finding to cyclical fluctuations in unemployment following [Shimer \(2012\)](#).⁴ [Shimer \(2012\)](#) assesses the relative contribution of each flow to unemployment-rate fluctuations, and [Elsby et al. \(2015a\)](#) and [Elsby et al. \(2019\)](#) apply a similar methodology to the participation rate. Most closely related, [Simmons \(2023\)](#) parses the employment-to-unemployment transition by reason in the SIPP and finds a larger contribution of separations to unemployment fluctuations; as discussed in the introduction, we differ in studying employment volatility and in adding the participation destination.

One of our central cyclical results, that laid-off workers become more attached to the labor force in downturns, connects to an established literature on cyclical attachment and the changing composition of the separation and unemployment pools. [Elsby et al. \(2015b\)](#) document that the unemployment pool skews toward more attached workers in recessions, and [Mueller \(2017\)](#) shows that the pool of separators shifts toward higher-pre-displacement-wage workers. Our measured

²A complementary literature focuses on jobs. For example, [Davis et al. \(2011\)](#) use the Job Openings and Labor Turnover Survey (JOLTS) to provide statistics on establishments' contributions to job destruction and creation.

³A non-exhaustive list of other papers using a CPS quit/layoff classification following [Flaim \(1973\)](#): [Flaim \(1969\)](#), [Schwab \(1974\)](#), [Deutermann Jr \(1977\)](#), [Bednarzik and Klein \(1977\)](#), [Job \(1979\)](#), [Freeman \(1980\)](#), [Gellner \(1975\)](#), and [Michaels \(2024\)](#).

⁴This relates to the unemployment-volatility puzzle ([Shimer \(2005\)](#), [Chodorow-Reich and Karabarbounis \(2016\)](#), [Hagedorn and Manovskii \(2008\)](#), [Ljungqvist and Sargent \(2017\)](#), and [Mitman and Rabinovich \(2019\)](#)).

countercyclicality of the share of layoffs flowing to non-participation is consistent with this evidence, and we present it as a complementary, higher-frequency measurement based directly on the reason for separation rather than as a new result. The mechanisms behind cyclical attachment, whether the destination split reflects labor supply, selection, or both; are studied structurally by [Krusell et al. \(2017\)](#) and [Cairó et al. \(2022\)](#), and are the subject of our companion paper [Ellieroth and Michaud \(2026\)](#). We do not adjudicate among these mechanisms here, and instead provide the well-measured moments on which such structural analysis can build.

Finally, our contrast with JOLTS in Section 7 draws on work reconciling household- and establishment-based flow measures, including [Faberman \(2005\)](#) and the establishment-size reweighting of [Davis et al. \(2011\)](#); direct employer-to-employer moves, which our series does not capture, are measured in the CPS by [Fujita et al. \(Forthcoming\)](#), though that flow cannot be split into quits and layoffs.

3 Data and Methodology

3.1 Data source

We use monthly data from the Current Population Survey (CPS) from January 1978 to December 2024. The CPS is a rotating panel survey of approximately 60,000 households, conducted by the US Bureau of Labor Statistics. While primarily designed for cross-sectional analysis, the CPS’s rotating panel structure allows us to match individuals across consecutive months, enabling the computation of month-to-month labor market transitions.

3.2 Methodology

We classify flows out of employment by reason for separation (quit or layoff) following methodology closely related to that used in the Integrated Public Use Microdata Series (IPUMS), and using the CPS questions first studied in [Flaim \(1969\)](#) and [Flaim \(1973\)](#). We further divide both quits and layoffs by workers’ participation choices after employment separation. An individual is considered unemployed if they do not have a job but report they are actively searching for a job, and considered a non-participant (or out of the labor force) if they do not have a job and are not seeking one. We study four distinct flows:⁵

- Employment to unemployment due to a quit (EUQ)
- Employment to unemployment due to a layoff (EUL)
- Employment to non-participation due to a quit (ENQ)
- Employment to non-participation due to a layoff (ENL)

⁵For a glossary of the naming convention of flows please see the Appendix A

Flows to Unemployment. Surveyors ask all unemployed individuals for the reason they became unemployed. Response options distinguish individuals who had lost jobs (due to temporary layoff, involuntary job loss, or ending of a temporary job), those who had quit jobs, those who were re-entering the labor force after an extended absence from the work force, and those who were seeking their first jobs (new entrants). We use these answers to classify a separation into unemployment as a layoff or quit as follows:

- Layoff: Job loser/on layoff, other job loser, temporary job ended
- Quit: Job leaver

This classification is readily available in the harmonized data in CPS IPUMS ([Flood et al. \(2023\)](#)) and constructed series of layoffs and quits to unemployment are provided by the Bureau of Labor Statistics (BLS). We do not use these series directly. Instead, we check that our series nearly replicate the series provided by the BLS as a check that our classification and harmonization procedures replicate the methods used by IPUMS. This ensures our newly constructed variables come as close as possible in methodology to the popular IPUMS variables researchers have historically studied.

Flows to Non-participation. The variable coding reason for leaving the last job is not available on CPS IPUMS for non-participants, nor are the constructed flow series of quits and layoffs to non-participation released by BLS. We instead work directly with a question asked to individuals to inquire their reason of non-participation. It has slightly changed over the years, but is a close variant of:

Why did ... leave that job?

We restrict our sample to anyone who has worked in the past 12 months to provide consistent measurement across time. Prior to 1994, surveyors asked this question to all non-participants who fulfill the following criteria: (1) currently not in the labor force, but worked for pay within the last five years, and (2) in the outgoing rotation group (ORG), which means the individuals are in month of sample 4 or 8. After 1994, the question is asked to individuals who (1) are currently not in the labor force, but worked for pay within the last 1 year, and (2) are in the outgoing rotation group (ORG).

The possible answer choices to the question have changed over time and we harmonize them as in [Ellieroth and Michaud \(2026\)](#). The answers across all months and years and define a layoff or quit as follows:

- Layoff: Temporary, seasonal or intermittent job completed, Slack work/business conditions
- Quit: Personal or family (including pregnancy), Return to school, Health, Unsatisfactory work arrangements

The question was not intended to be asked to retired or disabled individuals, however, a few of them name retirement as a reason and we exclude those. There are additional separations where the question asking the reason why the last job ended is not asked. These include, for example, retirements and disabled individuals, as they were supposed to be screened out. We label these separations as other, but the reader should think of them as “unknown” since these separations such as retirements can certainly be preceded by an involuntary layoff as well as a planned quit.

Linking over Time & Flow Series Construction We follow standard methods to compute gross flows and transition rates that have been developed in a long literature for a more coarse classification of flows that omit reason for an employment separation and just look at transitions from employment (E) to unemployment (U) or non-participation (N) (Abowd and Zellner (1985a), Shimer (2012), Elsby et al. (2015a), and many others). We employ established harmonizing and linking methods used by IPUMS in order to come as close as possible to the variables available in IPUMS that are heavily used by researchers in Economics and other fields. We specifically follow Madrian and Lefgren (1999) when linking individuals across two consecutive months and verify the quality of the matches based on sex and age.⁶ This method ensures that when we aggregate our flows to broad EN and EU rates, we recover the same transition probabilities that would be computed from IPUMS harmonized CPS data.

To compute aggregate flow rates, or probabilities, we simply divide the number of transitions by how many individuals are in each labor market state.⁷ An exception is layoffs and quits into non-participation because only individuals in the outgoing-rotation groups are asked about their reason for non-participation. This means there are only a subset of individuals responding to the question. We assume that the distribution of individuals by reason for non-participation is the same across all individuals in that month and use the share of quits and layoffs from the outgoing rotation groups multiplied by the total EN transition rate to compute the number of layoffs for all other individuals making an employment to non-participation transition.

Lastly, we seasonally adjust the data using the X-13ARIMA-SEATS seasonal adjustment program provided by the U.S. Census Bureau.

3.3 Robustness and External Validation

Rotation Group Bias It is well established that the CPS suffers from rotation group bias due to the structure of the survey and attrition (Krueger et al. (2017), Bernhardt et al. (2024) among others). The reason-for-separation question is asked only in the two outgoing rotation groups, month-in-sample (MIS) 4 and 8, which we pool throughout. This means that there is a possibility that individuals in the ORG are systematically different from everyone.⁸

⁶We are not matching based on race since the answers to this question have changed drastically over time.

⁷Consistent with best practices advise by IPUMS, we do not use weights in constructing the flow series.

⁸Note that this robustness check is only relevant to the quit/layoff distinction into non-participation as the question to unemployed individuals covers everyone.

First, we verify that the two rotations do not differ, i.e. we check that a leaver observed in MIS 4 does not differ systematically from one observed in MIS 8 in the reason reported or in observable characteristics. We test both margins on the matched EN leaver sample, separately for the pre- and post-1994 reason-question regimes and using both unweighted and survey-weighted estimates. Neither margin shows a meaningful difference in either era: the layoff share among coded leavers and the full covariate distribution (age, sex, education, occupation, industry) are statistically indifferent across MIS 4 and MIS 8, which justifies pooling the two rotations. Full methodology and results are reported in Appendix B.1.

A second concern is whether the outgoing rotation groups, the only ones asked the reason question, are representative of the broader set of separations. Our constructed series apply the ORG quit/layoff split to *all* EN separations, including the four rotations (MIS 2, 3, 6, 7) that are never asked why the respondent left. We therefore compare the observable characteristics of ORG leavers (MIS 4 and 8) with non-ORG leavers on age, sex, and education, separately by era and using both unweighted and survey-weighted estimates. The two groups are very similar: ORG leavers are younger by about a quarter of a year at most (0.13 years pre-1994, 0.27 post-1994), differ in female share by under one percentage point, and are statistically indistinguishable in education. A joint test of all characteristics does reject, but this reflects the very large samples (roughly 90,000 and 160,000 leavers) rather than any economically meaningful gap. Beyond comparing composition, we also directly reweight the ORG quit/layoff split to match the observable composition of all six rotations (specification M2 in Section B.4). Consistent with the negligible gaps documented here, this correction does not change any of our findings regarding the level or business cycle dynamics of quits and layoffs. We conclude that extrapolating the ORG reason split to the remaining rotations introduces negligible composition bias. Full results are reported in Appendix B.2.

Non-response Because the reason for separation is asked only in the outgoing rotation groups, and only of respondents who clear a brief screener, it is not recorded for every EN leaver; our baseline imputes the coded quit/layoff split onto those leavers. We therefore verify that the imputation is insensitive to who lacks a code. Two features make this reassuring. First, most non-response carries no information about the quit/layoff margin: it reflects the mechanical narrowing of the question’s universe at the 1994 redesign, respondents screened out as retired or disabled (not separators of the kind we study), and a small share of spurious employment records. Second, the genuine separations that go uncoded differ from coded leavers only modestly, and along observable dimensions—primarily sex and education, with no role for age once the retired and disabled are set aside. A cyclical pattern in non-response could in principle bias these correlations; we find none. Interacting the response propensity with the unemployment rate reveals no systematic cyclical relationship, and re-estimating the correction in rolling windows reproduces the static results (Tables 25–27), so time-varying selection does not affect the dynamics. Appendix B.3 reports the full decomposition and composition analysis. Reweighting the split to undo this selection leaves all level and business cycle results unchanged

(Section B.4), confirming that non-response, while sizable in volume, is inconsequential for the quit and layoff series.

DeNUNifying the Data A well-known concern with linked CPS data is that a nontrivial share of measured transitions between unemployment (U) and non-participation (N) are spurious, generated by classification or reporting error rather than genuine changes in labor-force status (Abowd and Zellner, 1985b; Poterba and Summers, 1986). The standard correction, applied by Elsbey et al. (2015c) among others, “irons out” single-month reversals: a one-month unemployment spell bracketed by non-participation is recoded to non-participation ($NUN \rightarrow NNN$), and, symmetrically, a one-month non participation spell bracketed by unemployment is recoded to unemployment ($UNU \rightarrow UUU$), the middle month being treated as a misclassification and reassigned to the surrounding state. We apply this “deNUNification” to our matched sample and recompute the key findings.

Table 28 compares the baseline series with their deNUNified counterparts. The correction leaves the results essentially unchanged: every average is identical to two decimal places, and every correlation with the unemployment rate moves only in the third or fourth decimal. As with our other robustness checks, accounting for spurious labor-force transitions affects neither the levels nor the business-cycle behavior of quits and layoffs.

Other. In addition to the above robustness checks, we also check whether the moving average smoother we apply to the data affects the results and the findings in the appendix (B.6) show it does not. We also look at different categories within the layoff category to show that the layoff results are driven by true layoffs.

Validity of Quit/Layoff Split. Is the quit/layoff distinction meaningful for non-participants and is the participation decision distinct for those who have been laid-off? These are two common questions. Past research has argued that the answers to these questions on the reasons nonparticipants left the labor force are informative about future labor force participation. Using the same questions in the CPS that we do, Deutermann Jr (1977) finds that nonparticipant prime age men who left their last job due to economic reasons or layoff are more likely to expect to return to the labor force within a year than those whose job ended for other reasons.⁹

Another indication that reason for separation to non-participation provides distinct information beyond unemployment versus non-participation status comes from a question that asks individuals about their primary activity in non-participation.¹⁰ The possible answers are disabled, ill, in school, taking care of house or family, and other. We group disabled and ill together and add a category of “unknown” for all individuals for whom the answer was missing.

⁹See Schwab (1974) for men age 58-63.

¹⁰The variable is called nilfact and available for everyone 15+ years of age on CPS IPUMS from 1994 onwards. For this analysis we will restrict our sample to the working-age population only because the original unharmonized variable only includes people up until 49 years of age since it’s not meant to include retired individuals.

Activity	Reason for Non-participation	
	Layoff	Quit
Disabled/Ill	6%	12%
School	8%	18%
Family	49%	52%
Other	33%	15%
Unknown	4%	3%

Table 1: **Primary Activity by Reason for Non-Participation**

Table 1 shows some meaningful differences in primary activity during non-participation by reason for job separation. For example, twice as many quits as layoffs (30% versus 14%) to non-participation are due to disability/illness or school, two activities that are often associated with prolonged periods of non-employment. This table also provides evidence that the state of non-participation itself is meaningful. The majority of both groups report specific, valid reasons they are not looking for a job. For example, a similar share of non-participants report that their main activity is family care regardless of whether they quit or were laid off.

3.4 External Validation: Panel Study of Income Dynamics (PSID)

We compare our series against the Panel Study of Income Dynamics (PSID) for external validation on the reliability of our CPS classification and imputation. The PSID is a longitudinal panel with questions that allow us to reconstruct the same reason-by-destination classification we built in the CPS. The PSID does have some short comings compared to the CPS. It is far smaller than the CPS, released with a year or more lag, and is a biennial retrospective survey over our sample period (2001-2019).¹¹ For these reasons, we will pool the PSID sample and use it to judge whether we are measuring the right average reason-cross-destination composition of separation, but it is not large enough to validate flow rates for specific months or business-cycle moments.

We construct PSID analogs of the four flows and restrict to prime-age 25–55, to match the sample from the CPS for which we have more complete responses and which we discuss in the main text. The mapping from PSID reason-for-separation categories to our quit/layoff definition is not one-to-one with the CPS questions; we describe the crosswalk in Appendix E.

The compositional shares computed in the PSID match closely those in the CPS. The right-hand block of Table 2 reports, for separations to non-employment, the destination split of quits and of layoffs in the CPS and the PSID. Among quits to non-employment, the CPS assigns a share 0.866 to non-participation against 0.878 in the PSID (a gap of 1.2 percentage points); among layoffs to non-employment, the CPS assigns 0.359 to non-participation against 0.377 in the PSID (1.8 percentage points). Every share in the full-sample comparison lies within 2 percentage points of its PSID counterpart. Given the independence of the two surveys, we read

¹¹These are the years for which we can construct a monthly panel of job separations similar to the CPS.

	Share of <i>all</i> separations (incl. E-to-E)			Share conditional on non-employment				
	PSID		SIPP	CPS		PSID		SIPP
	All	Rec.	All	All	Rec.	All	Rec.	All
<i>Panel A. Quits</i>								
to N (QN)	0.47	0.55	0.42	0.87	0.85	0.88	0.88	0.79
to U (QU)	0.07	0.08	0.11	0.13	0.15	0.12	0.12	0.21
to E (QE)	0.49	0.39	0.47	—	—	—	—	—
Total	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
<i>Panel B. Layoffs</i>								
to N (LN)	0.29	0.23	0.24	0.36	0.31	0.38	0.28	0.39
to U (LU)	0.48	0.59	0.37	0.64	0.69	0.62	0.72	0.61
to E (LE)	0.23	0.18	0.39	—	—	—	—	—
Total	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Table 2: Share of separations flowing to each destination: (N)on-participation, (U)nemployment, or directly to another (E)mmployer; by type of separation, (Q)uit or (L)ayoff. The left block reports each destination as a share of *all* separations of that type; the right block conditions on separations into non-employment. “Rec.” restricts to recession months. Columns sum to one within each panel. The CPS observes the quit/layoff distinction only for separations into non-employment and so appears in the right block only. See Appendix E Table 37 for PSID standard errors.

this as agreement consistent with sampling noise rather than systematic bias. Comparing the CPS point estimates to these standard errors confirms the reading: every CPS share in Table 37 lies within roughly 1.5 standard errors of its PSID counterpart, in both the full sample and the recession subsample, so that none of the eight comparisons is statistically distinguishable at conventional levels.

Despite greater noise from the smaller sample, the PSID independently reproduces the qualitative properties of quits, layoffs, and their destination over the cycle. In the CPS the share of layoffs flowing to non-participation is countercyclical: it falls from 0.359 over the full sample to 0.310 in recessions. The share is also countercyclical in the PSID: it falls from 0.377 to 0.284. Both surveys thus show laid-off workers becoming *more* likely to stay in the labor force during downturns. The quit-to-non-participation share likewise edges down in recessions in both surveys (0.866 $\rightarrow$ 0.854 in the CPS; 0.878 $\rightarrow$ 0.877 in the PSID). The recession-conditional differences in the datasets are larger than the full-sample differences, though still within sampling error (Appendix E reports the underlying cell counts and standard errors) and still a validation of the cyclical moments.

The PSID comparison is free from imputation and so serves specifically as a check on our imputation method in the CPS. In the CPS, the reason-for-leaving question is asked only of respondents in the outgoing rotation groups and we use their share to scale the full employment-to-non-participation flow, under the assumption that outgoing-rotation-group separators are

representative of all such separators. The PSID is not subject to that restriction: it observes the reason for all flows. The close agreement of the imputation-based CPS shares with the un-imputed PSID shares is therefore direct evidence that the outgoing-rotation-group representativeness assumption holds.

The left-hand *Total Flows* block of Table 2 is not a validation step and instead places our series in context. We are clear throughout this paper that the CPS construction observes only separations to non-employment and does not include quits or layoffs directly to another employer without a non-employment spell. The PSID, by contrast, includes the reason for all flows and shows that roughly half of all quits (48.8%) and about a quarter of all layoffs (23.0%) are direct employer-to-employer moves that our series, by design, does not capture. This is not a defect of the series but a statement of its scope, and partially reconciles our quit totals with total separations via quits reported by employers in the JOLTS.

3.5 External Validation: Survey of Income and Program Participation (SIPP)

The SIPP is a second, independent household survey from which we can reconstruct the same reason-by-destination flows, and it provides a secondary external check. On the layoff margin the SIPP corroborates the CPS closely: it assigns a share 0.389 of layoffs to non-participation against 0.359 in the CPS, and 0.611 to unemployment against 0.643. The quit margin diverges more; the SIPP assigns 0.209 of quits to unemployment against 0.133 in the CPS.¹² We therefore treat the SIPP as corroborating rather than definitive: the close agreement on the layoff-destination split is reassuring, while the wider quit gap plausibly reflects its substantially smaller sample relative to the CPS, rather than a deficiency of the CPS series.

4 Key Statistics for Macroeconomists

In this section we provide statistics on the incidence and cyclical variability of employment separations by reason and destination. We focus on statistics that we think would be of use to Macroeconomists. In particular, we provide statistics about total quits and layoffs resulting in non-employment. This is importantly distinct from prior work that established business cycle facts to validate a theory or calibrate/estimate a model using data on employment to unemployment flows only, or assuming employment to non-participation flows were all voluntary quits. We will show that the actual layoff and quit time series are quite different and should hold different implications both for quantitative analysis and economic theories.

All statistics and analyses in the main text are for the prime-age population, those between 25 and 55 years old, only. We replicate statistics and analyses for the working-age population, those 16 years and older, in the appendix. We focus on the prime-age population because retirees are not asked the reason for non-participation question. Retirements make up a sizable share

¹²Simmons (2023) studies similar flows in the SIPP but does not include the cyclicity of the destinations of quits and layoffs.

of EN flows and we do not want to impute these flows as quits or layoffs because our analysis of PSID data shows flows into retirement include both quits and layoffs and the share of each moves over the business cycle.

The series span January 1978 to December 2024 and are smoothed using a 6-month centered moving average smoother.¹³ This is a typical filtering choice for CPS data and we show in the appendix that alternative 3 or 4 month, centered or lagging, smoothing windows affect the series little except for the layoff series in times around extreme spikes such as COVID-19. During those times, shorter windows provide more volatile swings in layoffs as would be expected.

4.1 An Overview of Quits and Layoffs to Non-Employment

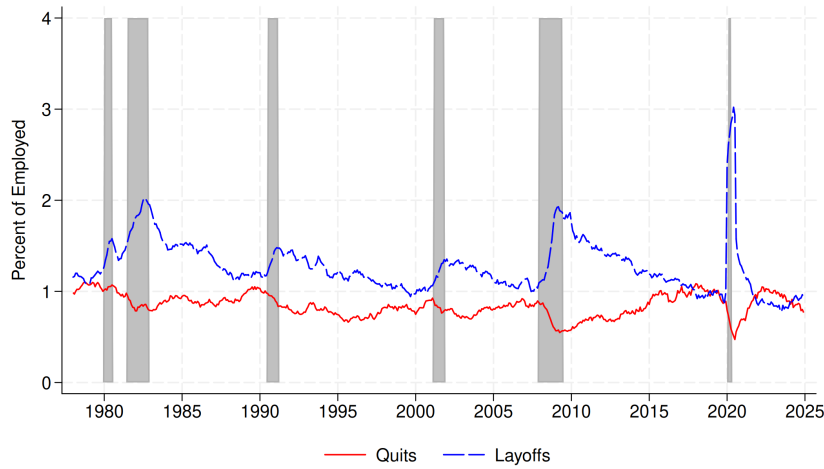


Figure 1: **Monthly Quit and Layoff Rates to Non-Employment** Prime Age. Shown as a percent of employment. Monthly seasonally-adjusted data and 6-month centered moving average.

Figure 1 plots the time series of total monthly quit and layoff rates to non-employment from 1978-2024 as a percent of total employment for the prime-age population (25-55 years old) in the United States. Table 3 provides averages for these series and their correlation. These quits and layoffs series combine separations to both unemployment and non-participation.

	Total	Quits	Layoffs	Corr(Q,L)
1978-2024	2.12	0.85	1.27	-0.46
1978-1990	2.35	0.94	1.41	-0.65
1991-2019,2021-2024	2.05	0.81	1.24	-0.41

Table 3: **Statistics on Monthly Quit and Layoff Rates to Non-Employment** Prime Age. Shown as a percent of employment. Monthly seasonally-adjusted data and 6-month centered moving average.

¹³In order to avoid any problems at the end of the time series due to smoothing using a centered moving average, we smooth the time series using data until mid-2025 and then restrict to end at 12/2024.

In most months, layoffs exceed quits as a share of separations to non-employment. On average, 40% of all separations to non-employment are quits and 60% are layoffs as an average of 1.27% of workers leave employment via layoff each month and 0.85% via a quit. There are some exceptions to this regularity. Notably, the quit rate exceeded the layoff rate during the tighter labor markets of the late 2010s and 2022-2023. Both types of separations became rarer after 1990. Although the extreme year of 2020 is excluded, it is not clear whether this decline is a feature of long cyclical dynamics over this period or of a more fundamental structural change.

The rates of quits and layoffs to non-employment are negatively correlated. When layoffs rise, quits fall. The correlation is -0.46 for the whole time series, and -0.65 during the pre-1990's period and -0.41 post-1990. The drop in the correlation is even stronger for the working age population as shown in Appendix Table 29.

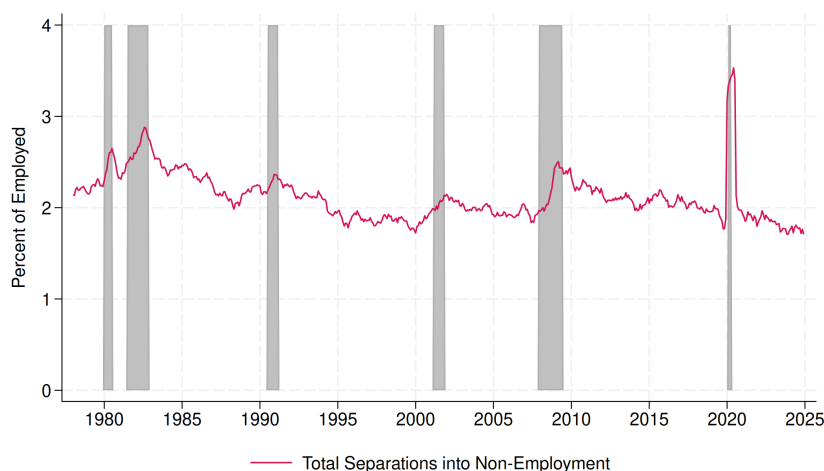


Figure 2: **Total Separations into Non-employment** Prime age. Total = Quits + Layoffs. Shown as a percent of employment. Monthly seasonally-adjusted 6-month centered moving average data.

Figure 2 shows the sum of the two series, i.e. total separations into non-employment. In the average month, 2.1% of workers leave their job to non-employment either as a result of a quit or a layoff. Although Figure 1 showed that both quits and layoffs vary with the business cycle, total separations move much less. The reason for that is that the decline in quits during recessions works to offset the rise in layoffs.

Figure 3 plots flows from employment to non-employment by destination: unemployment or non-participation. These are standard flows considered in prior research on the cyclical properties of the labor market, and yet they can be misleading if one assumes employment to non-participation flows to be voluntary and employment to unemployment flows to be involuntary. Comparing the EU and EN rates to our quits and layoffs series, we see that flows to non-participation exceed those to unemployment whereas layoffs exceed quits.

Table 4 shows how the EU and EN flows differ from quits and layoffs from employment shown in Table 3.¹⁴ Total EN flows are 1.8 times larger than quits to non-employment and layoffs are

¹⁴Total flows, the sum of EU and EN, are higher in Table 4 than total flows, the sum of quits and layoffs, in 3

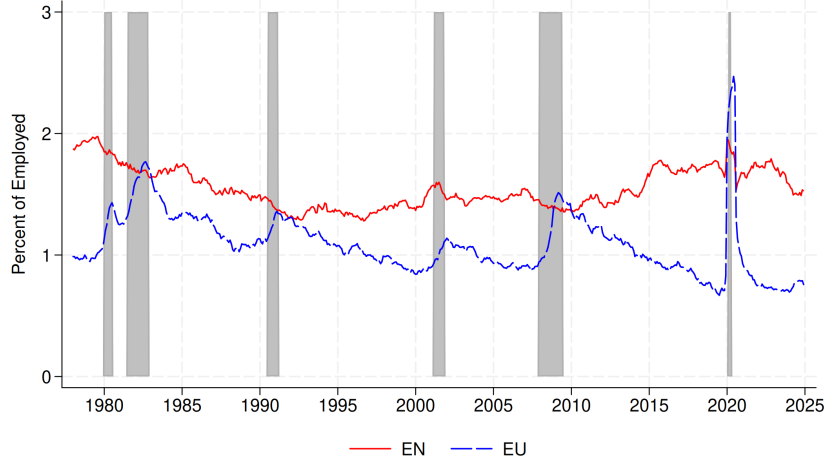


Figure 3: **Flow Rates from Employment to Non-Participation (EN) and to Unemployment (EU)** Prime Age. Shown as a percent of employment. Monthly seasonally-adjusted data and 6-month centered moving average.

	Total	EN	EU	Corr(EN,EU)
1978-2024	2.64	1.55	1.09	0.01
1978-1990	2.92	1.67	1.25	-0.13
1991-2019,2021-2024	2.53	1.51	1.03	-0.09

Table 4: **Separation Rate by Destination.** (Prime Age. Total = EN + EU. Shown as a percent of Employment. Monthly seasonally-adjusted 6-month centered moving average data)

1.2 times more common than EU flows. Furthermore, the negative correlation between quits and layoffs to non-employment is more stark. EN and EU flows show no correlation over 1978-2024 but quits and layoffs to non employment have a -0.46 correlation.

Comparing Figure 4 and Figure 1, it is obvious that quits are related to total flows to N and layoffs to total flows to U, but they are not the same. Notably, the rate of EN flows is around 80% higher at a monthly frequency than the quit rate. There are also clear cyclical differences, with the quits and layoffs each more volatile than EN and EU, respectively. Figures 4 and 5 allude to the reason for these discrepancies. A significant share of workers choose non-participation after a layoff, which contributes to EN flows being significantly larger than quits in the data. This share decreases during recessions and generates less volatile EN flows.

Table 5 shows an average of 34.8% of prime-age workers move to non-participation if they are laid off to non-employment. That means that they had a job last month, lost it, and decide not to look for a job this month. For workers who quit, the share who move to non-participation is much higher at 84.7%. This makes it clear that it should not be assumed that all movements to non-participation are due to a quit decision. Indeed, classifying flows from employment to non-participation because some EN and EU flows cannot be classified as quit or layoff or are misclassified. A nontrivial number of individuals who makes an EU transition reports (wrongly) being a re-entrant or new entrant into the labor force.

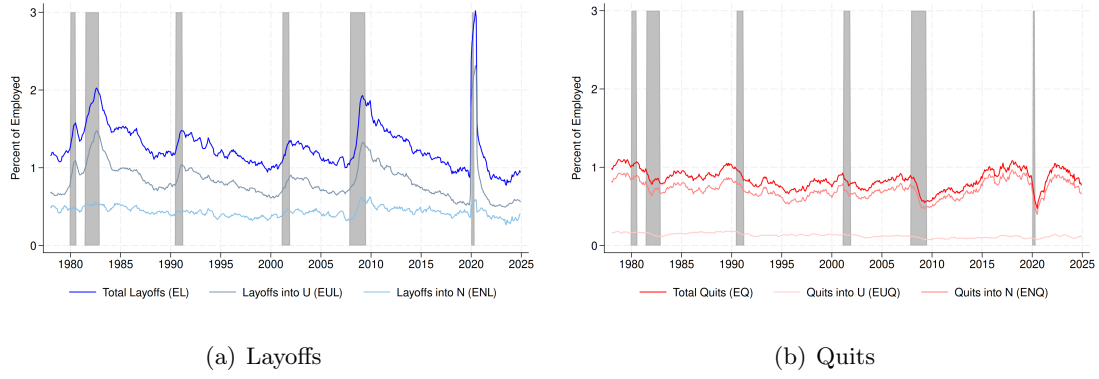


Figure 4: **Quit and Layoff Rates by Destination: Unemployment (U) or Non-participation (N).** (Prime Age. Shown as a percent of Employment. Monthly seasonally-adjusted 6-month centered moving average data)

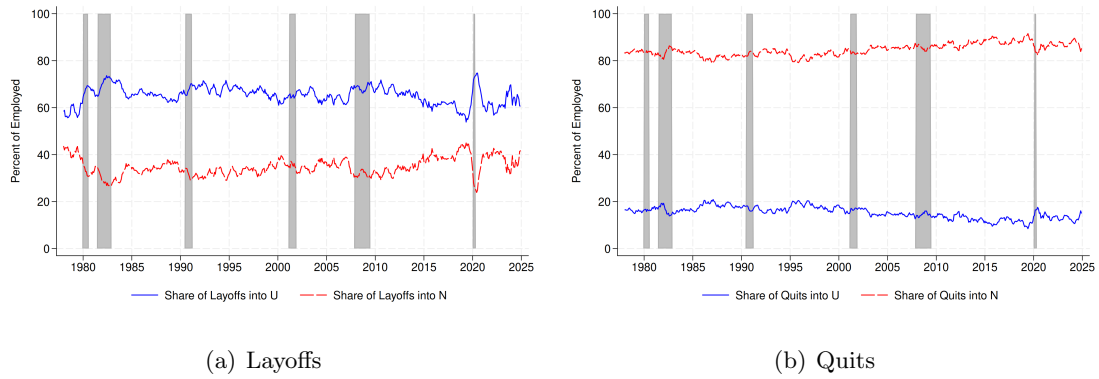


Figure 5: **Share of Each Separation to Unemployment or Non-Participation.** (Prime Age. Monthly seasonally-adjusted 6-month centered moving average data)

non-participation as quits understates the level of lay-offs by 40%, which will be especially important considering business cycle fluctuations.

	% Quits to		% Layoffs to		Corr(sQN,sLN)
	U	N	U	N	
1978-2024	15.3	84.7	65.2	34.8	0.38
1978-1990	17.2	82.8	65.9	34.1	-0.10
1991-2019,2021-2024	14.6	85.4	65.1	34.9	0.39

Table 5: **Percent of Quits and Layoffs to Non-Employment by Destination.** (sQN (sLN) = share of quits (layoffs) to non-participation. Prime Age. Monthly seasonally-adjusted 6-month centered moving average data)

Table 5 also shows that the share of workers choosing to look for a job after a layoff is stable across the early and late period of the sample. The share looking after a quit has decreased by 2.6 ppt from 17.2% to 14.6%. This does appear to be a secular decline despite business cycle

fluctuations. The correlation between the share of quit workers who exit the labor force and share of laid off workers who exit is positive on average but flips signs from the early to the late period. In the early period the relationship was negative and in the later it is positive. This finding is even stronger for the working age population as shown in Appendix Figure 31.

4.2 Business Cycle Statistics

Flow rates determine the business cycle properties of labor market stocks, such as the unemployment rate and employment-population ratios. Understanding their properties helps researchers understand how choices and frictions interact with each other and form over the business cycle.

We start by illustrating how quits and layoffs evolve over the business cycle. The first row of Table 6 shows the correlation of each flow rate (x) with the aggregate unemployment rate (y).¹⁵ Here we present results using the aggregate unemployment rate, defined as the number of unemployed divided by the total labor force, as the indicator of the cyclical state of the labor market (Foote et al. (2025)). In the appendix, we repeat this exercise using an alternative indicator of tight labor markets.

Statistic	Quits			Layoffs			Total sep.
	to U	to N	Total	to U	to N	Total	to (U+N)
$\text{Corr}(x, y)$	-0.16	-0.48	-0.46	0.82	0.75	0.84	0.72
$\text{SD}(x)/\text{SD}(y)$	0.02	0.06	0.09	0.14	0.04	0.17	0.15

Table 6: **Business Cycle Correlations: Quits and Layoffs.** (Prime age. Monthly. Correlations of each flow (x) with the unemployment rate (y))

Table 6 confirms that quits and layoffs move in opposite directions over the business cycle as shown graphically in Figure 1. Quits into both unemployment (U) and non-participation (N) decrease in recessions when unemployment is rising but quits into N decline much more strongly over the cycle than quits into U. The layoff column shows that layoffs respond much more to the business cycle as they rise during recessions when unemployment is rising. Layoffs into unemployment respond more than layoffs into non-participation during a recession. This difference is partially due to the finding that the share of laid-off workers who leave the labor force decreases in recessions.

Statistic	
$\text{Corr}(\text{Quits}, \text{Layoffs})$	-0.46
$\text{SD}(\text{Quits})/\text{SD}(\text{Layoffs})$	0.41

Table 7: **Business Cycle Relation of Quits and Layoffs** (Prime age. Monthly and seasonally adjusted.)

¹⁵None of the flow rate series show strong lower frequency trends, nor does the series of unemployment, and so we do not filter to remove trends.

Table 7 shows that the correlation between quits and layoffs is -0.46 over the entire sample period. This is starkly different from the correlation between EU and EN flows which is 0.01. Quits decrease in times when unemployment increases, and thus are procyclical. Layoffs move in the opposite direction, they are countercyclical and increase when the unemployment rate is high.

Statistic	Layoffs	EU	Quits	EN
$\text{Corr}(x, y)$	0.83	0.82	-0.46	-0.06
$\text{SD}(x)/\text{SD}(y)$	0.17	0.14	0.09	0.07

Table 8: **Business Cycle: Relationships with the Unemployment Rate (y)** (Prime age. Monthly and seasonally adjusted. Correlations of each flow (x) with the unemployment rate (y))

Table 8 shows the business cycle properties of EU and EN flow rates and compares them with the statistics for layoffs and quits which are taken from Table 6. The first two columns compare EU flow rates (those typically assumed to comprise all layoffs) with our layoff series. Both series are countercyclical with similar cyclical volatility. This is a lucky coincidence. When we compare the standard deviations of our layoff series with the standard deviations of the EU rate, we see that our layoff rate exceeds the EU rate in terms of volatility relative to the unemployment rate by over 20%. The researcher who considers layoffs to unemployment (a variable available in IPUMS) to comprise all layoffs would fare worse and find a cyclicity which is too low compared to the true cyclicity of total layoffs to both unemployment and non-participation.

The second two columns of Table 8 show starker differences between quits and EN rates. While our quit series is strongly procyclical, i.e. quits decline in recessions, the EN rate is only mildly procyclical. This means that the prior work on quits over the business cycle that considered all EN flows to be voluntary had the cyclical nature of voluntary quits wrong. An explanation for this difference is that around 40% of all EN transitions are actually precipitated by layoffs. We have seen in Table 6 that layoffs are strongly countercyclical and work to offset the procyclicality of quits into non-participation. We therefore strongly recommend using the quits series rather than the EN transition rate in order to analyze how labor supply choices, e.g. whether to quit from employment, shape employment and unemployment over the business cycle.

Statistic	Layoffs	EU	Quits	EN
$\text{Corr}(x, y)$	0.91	0.91	-0.05	0.36
$\text{SD}(x)/\text{SD}(y)$	1.13	0.98	0.46	0.63

Table 9: **Business Cycle: Relationships with the Total Separation Rate (y)** (Prime age. Monthly and seasonally adjusted. Correlations of each flow (x) with the total separation rate (y))

Table 9 repeats the exercise from the previous table but looks at the relationship with the

Recession	Total Separations in pp	Layoffs		EU	
		in pp	Relative	in pp	Relative
1980	0.41	0.35	0.85	0.37	0.90
1981	0.57	0.67	1.18	0.51	0.89
1990	0.21	0.27	1.59	0.19	0.90
2001	0.27	0.37	1.37	0.27	1.00
2007	0.67	0.92	1.37	0.62	0.93
2020	1.76	2.13	1.21	1.76	1.00

Table 10: **Recessionary increase in total separations, layoffs, and EU rate** Prime age, monthly. Peak-to-trough-change in percentage points (pp) and relative to total separations

total separation rate. Again, we see that both layoffs and the EU transition rate are similarly correlated with total separations. However, our layoff series shows a significantly higher volatility relative to the total separation rate than the EU rate. While the standard deviation of the EU rate is less or similar to that of the total separation rate, the standard deviation of our layoff series is significantly larger. Looking at Table 10, which shows the absolute and relative increase of layoffs and EU transitions in recessions, confirms the previous findings. Layoffs consistently fluctuate by more, i.e. increase by more during recessions, than total separations because of the dampening effect of quits.¹⁶ Our layoff series is comprised of layoffs into unemployment and non-participation, which both increase in recessions. The EU rate, however, only includes about 60% of all layoffs, those into unemployment, and in addition quits into unemployment which decline in recessions. Therefore, the EU rate significantly understates the volatility of layoffs in the economy.

5 Economic Significance

5.1 Importance of Destination after a Quit or Layoff

Table 11 shows how destinations, unemployment or non-participation, after quits and layoffs vary over the business cycle. The first main finding is that laid-off workers who separate from employment during recessions are more likely to flow to unemployment rather than non-participation. The second main finding is that workers quit less when unemployment is high. Taken together, in times when the unemployment rate is high, the share of each layoffs and quits into non-participation decline, as shown in Table 7.

The share of layoffs into non-participation is negatively correlated with the unemployment rate. This is important for understanding cyclical volatility in the labor market because the increase in the propensity of laid-off workers to remain in the labor force during a recession works to increase the cyclical volatility of unemployment more than the volatility of the layoff rate alone. In other words, when layoffs increase by a certain amount in a recession, flows into

¹⁶Table 9 shows that quits are negatively correlated with total separations.

Statistic	Share of Layoffs into N	Share of Quits into N
$\text{Corr}(x, y)$	-0.64	-0.15
$\text{SD}(x)/\text{SD}(y)$	0.02	0.02

Table 11: **Business Cycle: Relationship between Labor Force Exits (x) and the Unemployment Rate (y)** (Prime age. Monthly and seasonally adjusted.)

unemployment rise even more due to the increased propensity of laid-off workers to stay in the labor force.

The share of quits that flow into non-participation is also negatively correlated with the unemployment rate. We have seen the reason for that in Table 6: quits into non-participation strongly decline and by more than quits into unemployment during recessions, thus shifting the share of quits towards quits into unemployment. This is important for understanding cyclical volatility because the decrease in quits during recessions partially offsets the rise in layoffs and works to decrease the volatility of total separations from employment over the cycle.

Destination matters: How much do changes in the destination of laid-off workers amplifies cyclical unemployment volatility. Systematic cyclical changes in the destination of laid-off workers increases the cyclical variability of unemployment in a statistical sense. If the decision to report being unemployed versus a non-participant is superficial or has little meaning for actual job search efforts, then unemployment will be more volatile than actual slack. After all, the majority of hires from non-employment come from non-participation and so unemployment is hardly fully encompassing of the pool of potential workers. In this case, one might want to track an unemployment series that holds fixed the destinations of laid-off workers to better isolate the role of other contributing factors to unemployment dynamics.

We construct a counterfactual unemployment series that holds labor force participation patterns constant while allowing all other labor market flows to vary with the business cycle. The labor market consists of three stocks: employment (E_t), unemployment (U_t), and not in the labor force (N_t), that sum to the total population (normalized to one). These stocks evolve according to gross flows between them. Unemployment dynamics are given by

$$\Delta U_t = (EU_t + NU_t) - (UE_t + UN_t) \quad (1)$$

where EU_t denotes the flow from employment to unemployment, NU_t the flow from not in the labor force to unemployment, and so forth.

Gross flows are the product of hazard rates and source stocks. The EU flow depends on both quits and layoffs:

$$EU_t = \lambda_{eq,t}E_t + (1 - s_t)\lambda_{el,t}E_t \quad (2)$$

where $\lambda_{eq,t}$ is the quit rate to unemployment, $\lambda_{el,t}$ is the total layoff rate, and s_t is the share of layoffs flowing to not in the labor force rather than unemployment. Similarly, the employ-

ment to non-participation flow is $EN_t = \lambda_{en,t}E_t + s_t\lambda_{el,t}E_t$, where $\lambda_{en,t}$ captures quits to non-participation. Other flows follow analogously: $UE_t = \lambda_{ue,t}U_t$, $UN_t = \lambda_{un,t}U_t$, $NE_t = \lambda_{ne,t}N_t$, and $NU_t = \lambda_{nu,t}N_t$.

We consider a “participation neutral” series of unemployment that holds constant the flow rates between unemployment and non-participation as well as the share of workers laid off from employment who exit the labor force. To construct the participation neutral series, we hold three parameters constant at values $(s^*, \lambda_{nu}^*, \lambda_{un}^*)$ while allowing all other flow rates to vary over time. These values are chosen to maximize the correlation between actual unemployment U_t and counterfactual unemployment U_t^{cf} , subject to the constraint that all parameters lie in $[0, 1]$. We chose to maximize the correlation with actual unemployment rather than, say, to minimize counterfactual variance, because our aim is to strip out the cyclical destination of separations while distorting the level and comovement of the series as little as possible; a variance-minimizing choice would mechanically understate volatility for reasons unrelated to participation. Appendix F shows the volatility gap is essentially unchanged when the 2020 pandemic year is excluded, so the result is not an artifact of a single episode.

The counterfactual unemployment stock evolves according to

$$U_t^{cf} = U_{t-1}^{cf} + \left[\lambda_{eq,t}E_{t-1}^{cf} + (1 - s^*)\lambda_{el,t}E_{t-1}^{cf} + \lambda_{nu}^*N_{t-1}^{cf} \right] - \left[\lambda_{ue,t}U_{t-1}^{cf} + \lambda_{un}^*U_{t-1}^{cf} \right], \quad (3)$$

with initial condition $U_0^{cf} = U_0$. After the construction of the entire participation neutral series in this way, we then add a constant to the series so that it has the same mean as the actual unemployment/population ratio. The interpretation is: by fixing the labor force participation margins $(s^*, \lambda_{nu}^*, \lambda_{un}^*)$ while allowing cyclical flows to vary, this counterfactual isolates unemployment fluctuations that would occur if exits after layoff and flows between U and N remained constant throughout the business cycle.

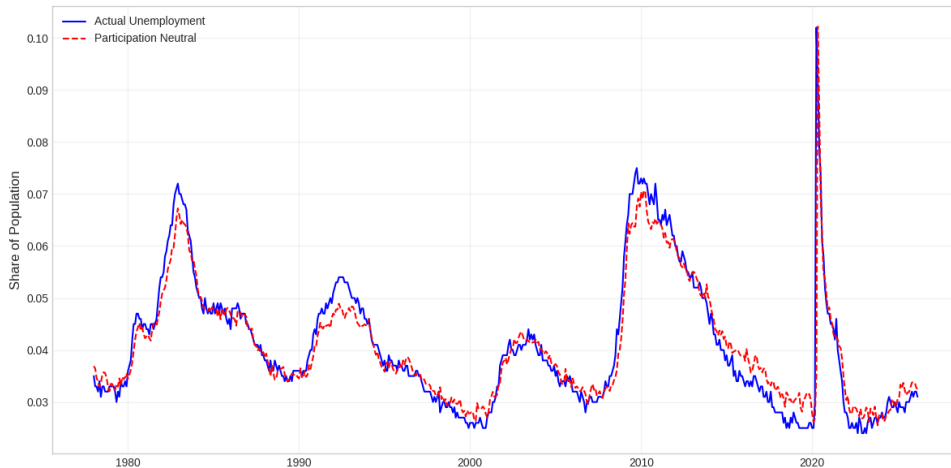


Figure 6: **Impact of participation flows on the dynamics of unemployment.** Participation neutral unemployment holds fixed each share of layoffs to nonparticipation, UN, and NU at values to maximize the correlation of the series.

	Actual	Partic. Neutral	Actual 6mo MA	Partic. Neutral 6mo MA
Std. deviation (monthly)	0.0127	0.0108	0.0123	0.0103
Corr. with GDP cycle (qtly)	-0.2360	-0.1896	-0.2506	-0.1988
Corr(Partic. Neutral, Actual)		0.9520		0.9833

GDP cycle: HP filter ($\lambda = 1600$). 6mo MA: 6-month centered moving average.

Table 12: Business-cycle statistics: actual unemployment vs. participation neutral, 1978–2025.

The results are plotted in Figure 6. The blue line is the actual series and the red-dashed line is the participation neutral series. Table 12 provides summary statistics of the series. The correlation between the two series is high at 0.95, but the actual series is more volatile than the participation neutral series. The actual series has a standard deviation of 0.0127, while the participation neutral series has a standard deviation of 0.0108. The higher volatility of the actual series implies that cyclical variation in participation outcomes amplify the volatility of unemployment by about 18%. When they are removed, we get the less volatile red-dashed line. Finally, the actual series is more highly correlated with GDP. Appendix F shows these patterns hold when we exclude the 2020 pandemic year, the volatility of the unemployment/population ratio is still 20% larger than the participation neutral unemployment/population ratio.

Period	Actual U	Participation-Neutral U	Relative Difference (%)
1979-1982	4.20	3.48	20.8
1990-1992	1.90	1.43	32.8
2000-2003	1.90	1.72	10.5
2007-2009	4.50	3.84	17.3
2019-2020	7.70	7.58	1.5

Table 13: **Unemployment/Population Changes During Recessions** First two columns: percentage point change from period min to max. Last column: difference between column one and two as a percent of column two.

The contribution of participation outcomes to unemployment volatility varies across recessions. Table 13 presents the changes in actual unemployment-to-population and in our constructed participation-neutral unemployment-to-population from min to max for each relevant period. The periods are chosen to capture the low of the unemployment rate preceding each recession and the maximum of the unemployment rate during the recession. The final column shows how much larger the increase in the actual series is relative to the increase in the participation neutral series, in percent terms.

The 1990’s recession stands out as having the largest difference between the actual unemployment and participation neutral unemployment. This is both because an unusually large amount of laid-off workers remained in the labor force and unemployed workers were less likely to exit to non-participation. These changes in destination caused unemployment to rise by

almost a third more than it otherwise would have. The COVID recession is also an extreme outlier. Although the share of laid-off workers leaving the labor force plummeted to record lows, this was mitigated by a record spike in the UN rate possibly due to misclassifications of workers awaiting recall (Hall and Kudlyak (2022)). The 2001 recession is another example where changes in participation outcomes did relatively little to exacerbate the rise in unemployment, adding only an additional 10.5% to the percentage point increase.

This colors how we compare volatility across recessions. For example, the rise in actual unemployment-to-population is identical in the 1990's and 2001 recessions but the participation neutral rise is 20% higher in the 2001 recession. This suggests more different dynamics in layoff rates and job finding rates across the two recessions than can be seen once we hold dynamics of participation outcomes to be fixed.

A separate but interesting feature seen in Figure 6 is that the actual unemployment series often leads our counterfactual series, particularly at the onset of a recession. This suggests that participation outcomes shift in anticipation of the cycle. Specifically, the share of laid off workers who go to unemployment and the net flows from N to U rise, increasing the unemployment stock before the job loss rate rises and the job finding rate falls. The very recent movements from 2023-on show a stark example of this lagged relationship. The actual rise in unemployment is first driven by a rise in net flows to unemployment from non-participation and after a layoff. This rise is completely absent from the participation neutral series until a year later when job finding starts to slow down and contribute to higher unemployment.

5.2 Shimer Decomposition

In this section, we follow the methodology of Shimer (2012) to evaluate the quantitative contributions of layoffs and quits to the dynamics of unemployment and employment rates over the business cycle. As in Shimer (2012), we start by computing (discrete) monthly flow rates as described in Section 3. We follow Shimer (2012) as closely as possible, with the only difference being that we split transitions into unemployment and non-participation by reason. The flows we study are illustrated in Table 14.

	From			To		
	E	U	N	E	U	N
E	f_{EE}	$f_{EUQ} + f_{EUL}$	$f_{ENQ} + f_{ENL}$			
U	f_{UE}	f_{UU}	f_{UN}			
N	f_{NE}	f_{NU}	f_{NN}			

Table 14: **Flow rates by source (left column) to destination (top row)**

We compute instantaneous transition rates based on the above flow rates to correct for time aggregation bias generated by individuals making multiple labor market transitions within a month. Next, we take quarterly averages of the monthly time series and smooth the data. We will start by explaining how we change the time aggregation method by Shimer (2012) to

account for the addition of reason for movement into non-employment from employment.

Time Aggregation Bias We construct instantaneous transition rates λ^{AB} which denote a movement from state $A \in \{E, U, N\}$ to state $B \neq A$ following the methodology of [Shimer \(2012\)](#). This gives us six transition rates for each month: EU, EN, UE, UN, NE, NU. We use the two rates from employment to non-employment, λ^{EU} and λ^{EN} , to calculate the four rates from employment to non-employment by reason. In order to do that, we multiply the instantaneous transition rate λ^{EU} and λ^{EN} by the fraction of workers who quit or were laid off for every month in the observed time period to construct the four flow rates by reason and destination: λ^{EUQ} , λ^{EUL} , λ^{ENQ} , and λ^{ENL} .¹⁷

Decomposition We start by assuming the economy is in steady state which requires the flows in and out of employment and in and out of unemployment to be equal. Mathematically, this is:

$$\underbrace{(\lambda^{EUQ} + \lambda^{EUL} + \lambda^{ENQ} + \lambda^{ENL})e}_{\text{flows out of employment}} = \underbrace{(\lambda^{UE}u + \lambda^{NE}n)}_{\text{flows into employment}} \quad (4)$$

$$\underbrace{(\lambda^{UE} + \lambda^{UN})u}_{\text{flows out of unemployment}} = \underbrace{(\lambda^{EUQ} + \lambda^{EUL})e + \lambda^{NU}n}_{\text{flows into unemployment}} \quad (5)$$

where λ^{AB} indicates the instantaneous transition rates from A to B. We also assume that all individuals can be classified as either employed (e), unemployed (u) or not in the labor force (n):

$$e + u + n = 1 \quad (6)$$

Equation 4 is modified to include our quit and layoff series. Instead of looking at outflows by destination, we use our quit and layoff series to group outflows of employment by reason and destination. Again, we try to remain as similar as possible to the original approach in that the only modification we make is to replace λ^{EU} with $\lambda^{EUQ} + \lambda^{EUL}$ and λ^{EN} with $\lambda^{ENQ} + \lambda^{ENL}$.

Solving equations 4, 5, and 6 for e , u , and n , we get

$$e = k (\lambda^{UE}\lambda^{NU} + \lambda^{NE}\lambda^{UE} + \lambda^{NE}\lambda^{UN}) \quad (7)$$

$$u = k ((\lambda^{EUQ} + \lambda^{EUL})\lambda^{NE} + (\lambda^{EUQ} + \lambda^{EUL} + \lambda^{ENQ} + \lambda^{ENL})\lambda^{NU}) \quad (8)$$

$$n = k (\lambda^{UE}(\lambda^{ENQ} + \lambda^{ENL}) + \lambda^{UN}(\lambda^{EUQ} + \lambda^{EUL} + \lambda^{ENQ} + \lambda^{ENL})) \quad (9)$$

where k is a constant.

We will use these flow-based equations to calculate the contribution of each of the ten flow rates to the unemployment rate $\frac{u}{e+u}$ and the employment-to-population ratio $\frac{e}{e+u+n}$ each month.

¹⁷This approach relies on the assumption that quits and layoffs into U and N have a similar time bias in each month. While we cannot directly test for it, this approach allows us to calculate the instantaneous transition rates similar to the original paper by [Shimer \(2012\)](#) with minimal additional assumptions or restrictions.

Contribution of each Flow to the Unemployment Rate. We start by taking quarterly averages of the monthly data and smooth the time series using a 2-quarter-centered moving average. In order to quantify the effect of each flow rate, we compute alternative unemployment rates. We construct the time series of each alternative unemployment rate by setting all flow rates to their average value and only letting the flow rate of interest vary over time. We then regress the alternative unemployment rate on the unemployment rate constructed using the flow-based stocks.

Rate	Shimer (2012)	This paper	This paper
	1987-2010	1987-2010	1978-2024
	(1)	(2)	(3)
λ^{EUQ}	-	-0.04	-0.02
λ^{EUL}	-	0.28	0.41
λ^{ENQ}	-	-0.04	-0.05
λ^{ENL}	-	0.04	0.04
λ^{EU}	0.22	-	-
λ^{EN}	-0.05	-	-
λ^{UE}	0.51	0.54	0.42
λ^{UN}	0.16	0.11	0.11
λ^{NE}	0.08	0.05	0.06
λ^{NU}	0.13	0.09	0.10

Table 15: **Decomposition of fluctuations in the unemployment rate: All Flows**

Rate	Shimer (2012)	This paper	This paper
	1987-2010	1987-2010	1978-2024
	(1)	(2)	(3)
λ^{EU}	0.22	0.24	0.39
λ^{EN}	-0.05	0.00	-0.01
λ^{EL}	-	0.32	0.45
λ^{EQ}	-	-0.08	-0.07

Table 16: **Decomposition of fluctuations in the unemployment rate: Quits and Layoffs vs. EU and EN**

Table 15 shows the quantitative importance of each instantaneous flow rate to the fluctuation of the unemployment rate, and Table 16 summarizes the differences between our total quits and layoffs series compared to the standard EU and EN transition rates.

Columns (1) and (2) show the results when we restrict our data to cover the same time period as in Shimer (2012). Similar to Shimer (2012), we find that fluctuations in the job finding rate from unemployment, UE, contribute the most to fluctuations in the unemployment rate (regression coefficient 0.54 our data vs. 0.51 original data). To support our methodology, Table 16 shows the contribution of λ^{EU} computed by adding up λ^{EUQ} and λ^{EUL} is similar

to what Shimer (2012) finds.¹⁸ Additionally controlling for the reason of separation, however, reveals that the contribution of the layoff rate, EL, is about 35% larger than the EU rate.¹⁹ Thus, we show that the layoff rate, EL, contributes significantly more than would be assumed by only looking at the EU rate. The reason is that the rise in the unemployment rate due to the rise in the layoff rate is partially offset by the decline in quits EQ.²⁰

Another result is that movements within non-employment, those between unemployment and non-participation, account for about one third less of the cyclical changes in unemployment once we include our quit and layoff distinction. This reinforces our claim that the reason for a separation carries additional information above the destination that is important for understanding fluctuations in the unemployment rate.

Results for the entire 1978-2024 time period (Column (3)) show that UE rates have not been the dominant source of movements in unemployment outside of the 1987-2010 period. Instead, layoffs become the largest contributor to the fluctuations in the unemployment rate when we expand the sample to include the early 1980s recession, more of the 2010's recovery, and the pandemic. However, most of this finding is driven by the pandemic recession, where layoffs experienced historically high levels.

Figure 7 shows the actual unemployment rate (UR); the flow-based unemployment rate if only our layoff series varies (Counterfactual UR: Layoffs); and the flow-based unemployment rate if only the UE rate varies (Counterfactual UR: UE). Although layoffs and UE are the main contributors to the fluctuations in the unemployment rate from our Shimer decomposition, we can see from the figure that the importance of each of the two flow rates differs at different points in the business cycle. Fluctuations in our layoff series are more important to fluctuations in the unemployment rate at the beginning of a recession whereas fluctuations in the UE transition rate contribute more towards the end of a recession and during expansionary periods. It appears that layoffs are more important at turning points, whereas the job finding rate from unemployment contributes more to the persistence of fluctuations in the unemployment rate.

In addition, the figure helps to understand why layoffs are the most important contributor to the volatility of the unemployment rate over the entire time period; the pandemic recession was characterized by an increase in layoffs which outweighed the contribution of the UE transition rate.

Contribution of each Flow to the Employment-Population Ratio. We decompose the drivers of fluctuations in the employment-population ratio in an analogous fashion as we did for the unemployment rate. Tables 17 and 18 show our findings, the importance of different

¹⁸The difference is due to EU and EN flows that we cannot classify as quit or layoff and subsequently drop from this analysis.

¹⁹The total contribution of layoffs is $\lambda^{EUL} + \lambda^{ENL} = 0.28 + 0.04 = 0.32$ vs 0.24

²⁰This result is of similar spirit to Simmons (2023). He finds the contribution of separation rates to cyclical unemployment fluctuations to be over five times as large when accounting for reason for quit using SIPP data. We find the overall contribution of separations to be larger in the CPS data at 32% versus his finding of 25.4%, and we show the contribution is even larger for Employment-Population at 49%.

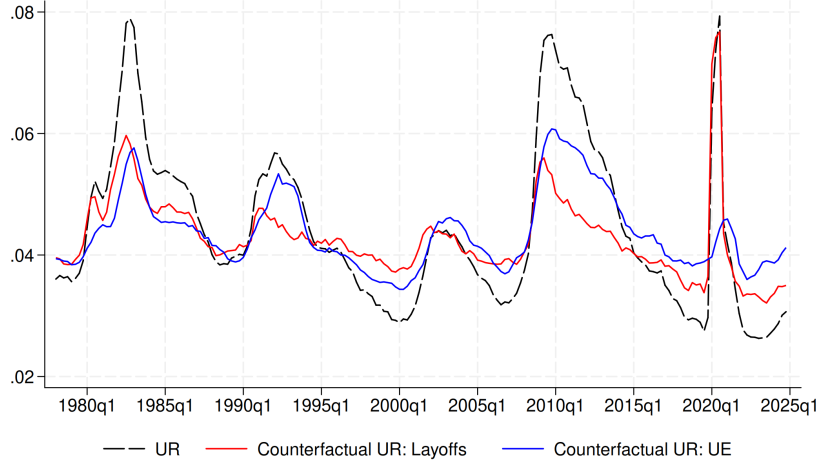


Figure 7: **Unemployment Rate and Counterfactual Unemployment Rate based on contribution of layoffs and UE transition rate** UR = Unemployment Rate. We construct the time series of each counterfactual unemployment rate by setting all flow rates to their average value and only let the the flow rate of interest (here: layoffs or EU) vary over time.

transition rates to fluctuations in the employment-population ratio.

Rate	Shimer (2012) 1987-2010	This paper 1987-2010	This paper 1978-2024
λ^{EUQ}	-	-0.04	-0.01
λ^{EUL}	-	0.43	0.44
λ^{ENQ}	-	-0.14	-0.01
λ^{ENL}	-	0.24	0.18
λ^{EU}	0.30	-	-
λ^{EN}	-0.40	-	-
λ^{UE}	1.16	0.93	0.50
λ^{UN}	-0.50	-0.24	-0.14
λ^{NE}	0.73	0.26	0.27
λ^{NU}	-0.42	-0.18	-0.05

Table 17: **Decomposition of fluctuations in the employment-to-population ratio: All Flows.**

The contribution of layoffs to fluctuations in the employment-population ratio is about 70% larger than just looking at the EU transition rate would suggest, as shown in Table 18. The reason is twofold. First, quits into unemployment fall in recessions and offset some of the effect of increased layoffs into unemployment. Second, and more importantly, layoffs into non-participation, λ^{ENL} , have a significant impact on the fluctuations in the employment-population ratio, but are not captured when just examining flows from E to U. Two-thirds of the importance of layoffs to the employment-population ratio stems from fluctuations in layoffs into unemployment and the other third is due to fluctuations in layoffs in non-participation.

Rate	Shimer (2012) 1987-2010	This paper 1987-2010	This paper 1978-2024
λ^{EU}	0.30	0.39	0.43
λ^{EN}	-0.40	0.10	0.17
λ^{EL}	-	0.67	0.62
λ^{EQ}	-	-0.18	-0.02

Table 18: **Decomposition of fluctuations in the employment-to-population ratio: Quits and Layoffs vs. EU and EN**

This is why fluctuations in layoffs account for a larger portion of the decline of the employment-population ratio in recessions than the rise in unemployment.

Table 18 also shows that quits, λ^{EQ} , had a dampening effect on employment volatility for the 1987-2010 period. This is because quits are negatively correlated with the unemployment rate, a fact explored in recent research on the importance of quits to business cycle fluctuations.²¹ A comparison with Table 16 shows that quits have a larger dampening effect on fluctuations in the employment-population ratio than they do on fluctuations in the unemployment rate. This is because quits account for a much larger share of flows to nonparticipation than to unemployment.

6 Differences and Similarities Across Recessions and Expansions

In the previous section we analyzed the different flow rates' contributions to overall business cycle fluctuations in the labor market. This section shows that our findings are robust to the different economic downturns and expansionary periods in our time frame.

6.1 Recessions

Our sample covers six NBER dated recessions starting with the 1980 recession up until the pandemic recession of 2020. Although recessions differ in their causes, lengths, and impacts on the labor market, we will show in this section that they are remarkably similar with regards to dynamics of quits, layoffs, and total separations into non-employment. Thus, we do not think that our empirical findings are the result of specific recessions having a dominating effect on the overall time-series.

The length of our series lets us ask whether the cyclical relationships documented above are stable features of the business cycle or particular to individual episodes. Our sample covers six NBER-dated recessions, from the 1980 recession through the pandemic recession of 2020, and we find that the composition of separations is strikingly consistent across all of these episodes. This is a new finding because no prior dataset with reason-for-separation could go back so

²¹See e.g. Ellieroth and Michaud (2024)

far; quits and layoffs in the JOLTS and the SIPP begin only around 2000 and the mid-1990s, respectively.

Recession	UR	EPR	Quits	Layoffs
1980	1.90	-1.20	-0.04	0.35
1981	3.60	-1.80	-0.19	0.67
1990	2.60	-1.40	-0.23	0.27
2001	2.40	-2.10	-0.16	0.37
2007	5.30	-4.40	-0.35	0.92
2020	9.70	-8.40	-0.42	2.13

Table 19: **Comparing Recessions.** Peak-to-trough change in percentage points. UR = Unemployment Rate. EPR = Employment-to-Population Ratio. Prime-age population

Table 19 shows that layoffs rise in every recession, and quits fall in every recession. This means that total separations move less over the cycle than layoffs alone as the decline in quits during most recessions partially offsets the rise in layoffs. The resulting change in total separations varies in size and depends on the relative sizes of the decline in quits compared to the increase in layoffs in an episode. The recessions listed in the table vary in length and in intensity, as can be seen by the differing increases in the unemployment rate.

We use labor market tightness as a measure of the state of the economy to better compare flows across recessions of different magnitudes. Labor market tightness is defined as the vacancy rate in the economy divided by the share of the working-age population that is unemployed.²² Analyzing our time-series in relation to market tightness allows for a more equal comparison among very different recessions.

Figure 8 shows the relationship between labor market tightness and our quit series for each recession plus six months of recovery. Each set of colored dots represents a recession; we group the two recessions in the early 1980s because of the small gap between them. The slopes are generally positive: quits are low when tightness is low and rise as the labor market tightens. This holds for all recessions except 2001. The correlation between our quit series and tightness is high in every recession, ranging from 0.7 to 0.95. Beyond the within-recession relationship, the figure shows a shift over time: quits are lower relative to tightness in the 2000s recessions than in those of the 1980s and 1990s. Fewer quits out of employment for a given tightness could reflect decreased labor-market dynamism (Davis and Haltiwanger, 1999) or higher labor-force attachment (Ellieroth and Michaud, 2024).

Figure 9 shows the relationship between layoffs into non-employment and labor market tightness for the same recessions. The relationship is strongly negative in every recession: layoffs are high when tightness is low and vice versa. All recessions are characterized by a very similar slope, and the correlation between layoffs and tightness is high and within the narrow range of

²²We use data on the vacancy rate in the U.S. economy from Petrosky-Nadeau and Zhang (2021). They combine vacancy data from different sources and provide a series that spans most of the time frame of our data (1/1978-12/2017)

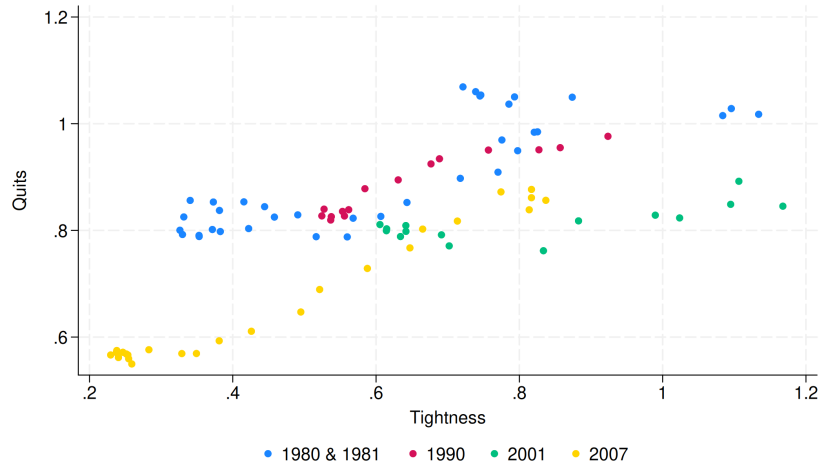


Figure 8: Comparing Recessions: Quits versus Labor Market Tightness Quits are the total quit rate from employed to non-employment. Market tightness is vacancies divided by the share of the working age population that is unemployed. Each recession includes the NBER recession months plus six months into the recovery. Data is for the working-age population.

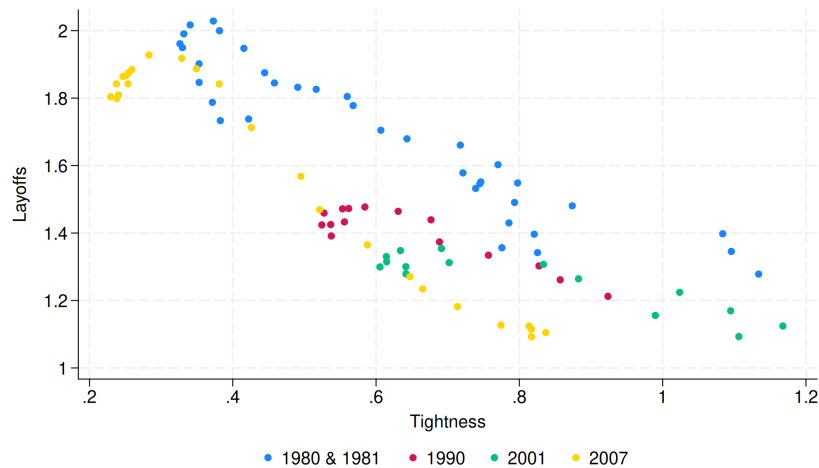


Figure 9: Comparing Recessions: Layoffs versus Labor Market Tightness Total layoffs are the total layoff rate from employed to non-employment. Market tightness is vacancies divided by unemployed. Each recession includes the NBER recession months plus six months into the recovery. Data is for the working-age population

0.90 to 0.94. What differs across recessions is the level. For a given degree of tightness, layoffs are markedly higher in the early-1980s recessions than in those of 1990, 2001, and 2007, which sit lower and closer together. This downward shift in the layoff–tightness relation is the layoff-side counterpart to the decline in quits per unit of tightness documented above. For a given state of the labor market, the earliest recessions in our sample delivered more total separations out of employment than later ones did. Such a secular decline in separations for a given degree of slack is consistent with the falling labor-market dynamism documented over this period ([Davis](#)

and Haltiwanger, 1999), and it is a pattern only a series spanning four decades of recessions can detect.

7 Context: A Comparison to the JOLTS

The Job Openings and Labor Turnover Survey (JOLTS) is a monthly employer survey run by the Bureau of Labor Statistics (BLS) and has been the primary source used to analyze quits and layoffs in the United States in near real-time.²³ Since the JOLTS are the series most researchers have used to study quits and layoffs over the cycle, we find it useful to explain the conceptual differences between it and our CPS series. This is not a validation exercise. The JOLTS measures a different object from our series, and the two should not be expected to agree; and where they diverge provides information about how the two concepts differ, not about whether either is correctly measured. Our aim in this section is to make those conceptual differences clear.

The most consequential difference is that JOLTS is broader: it counts quits and layoffs that move a worker directly from one employer to another, with no intervening non-employment spell, whereas we observe the quit-versus-layoff distinction only for separations into non-employment. Direct employer-to-employer moves are observed in the CPS, Fujita et al. (Forthcoming) recently improved the measurement of these flow but no question exists that can distinguish an E-to-E move as a quit or a layoff. While it is tempting to equate direct employer-to-employer moves with quits, our PSID evidence makes clear this is not a valid assumption. Table 2 shows that roughly a quarter of employer-to-employer separations are layoffs and that this share rises to about a third during recessions²⁴ The bottom line is that one cannot construct a comparable quit and layoff series from the CPS: the JOLTS quits correspond to our quits into non-employment *plus* employer-to-employer quits that we do not observe, and JOLTS layoffs correspond to our layoffs into non-employment *plus* employer-to-employer layoffs that we do not observe. Neither series is mismeasured; they count different sets of separations.

Two further features of JOLTS shape the comparison. First, JOLTS is known to underestimate separations even when sampling weights are applied, because it does not measure separations due to firm exit (Faberman, 2005); this problem is particularly acute when the economy is entering a recession. To remedy it, the disseminated JOLTS data are adjusted via a Monthly Alignment Method to produce stocks consistent with employment measured in the Current Employment Statistics (CES) (Cheng et al., 2009). These adjustments require assumptions about the composition of quits versus layoffs in the missing data. Second, JOLTS defines a quit more narrowly than we do. It defines layoffs as involuntary separations initiated by the employer, similar to our measure, but records retirements and transfers to other locations not as quits but under a separate “Other Separations” category, which also absorbs deaths and sep-

²³Other complementary and timely data sources include the Survey of Consumer Finance (SCF) as for example in Koşar and Van der Klaauw (2023).

²⁴Note that the PSID is less subject to time aggregation as we can see weekly diaries so these are not simply layoffs with short non-employment spells.

arations due to disability. Our reason-for-leaving measure also omits deaths but treats some of these differently, so even the non-employment portions of the two quit concepts are not defined identically.

For illustrative purposes, we provide a series that brings the CPS concept as close to the JOLTS concept as possible. As in JOLTS, we count an individual as laid off only if the job was lost involuntarily. For quits, we exclude the retired; disabled individuals are excluded automatically, because they are not in the universe asked the reason-for-non-participation question. Death is excluded automatically by our linking strategy, since a deceased person does not appear in the current month, and, because we consider only separations into non-employment, transfers to other locations do not arise. The earliest available JOLTS data are for January 2001, so we begin the comparison series then. Both series are seasonally adjusted.²⁵

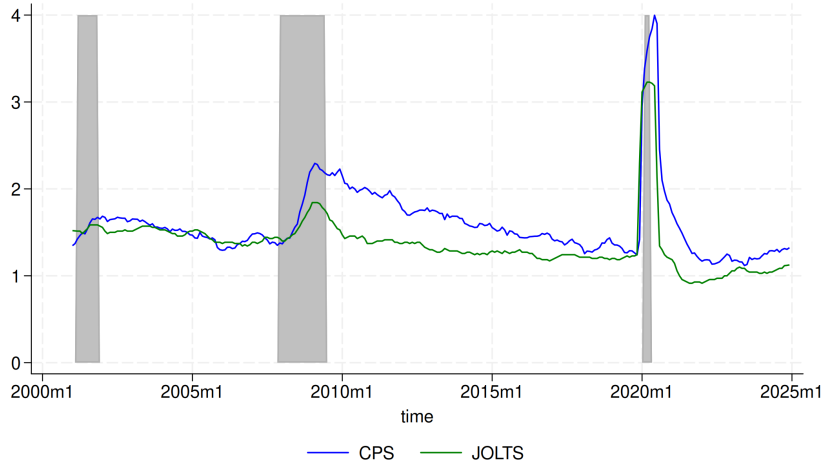


Figure 10: **Layoffs: CPS versus JOLTS**

Figure 10 compares the JOLTS layoffs series with ours. The correlation between the two over the full period is 0.60, a substantial departure from one, and our series is much more closely related to the unemployment rate: the correlation of our CPS layoffs with unemployment is 0.82, against only 0.29 for JOLTS. The two series track each other closely until the 2007 recession, after which our series exceeds JOLTS and the two slowly converge over 2012–2019; they deviate again during the pandemic recession and then converge more quickly. Two conceptual differences account for this pattern. Mechanically, JOLTS layoffs include the roughly one-quarter of layoffs that move directly to another employer, which our series structurally excludes, so a level gap is expected rather than a sign of error. And because our series counts only layoffs that end in non-employment, it is a product of both the layoff rate and the job-finding rate; this amplifies its cyclical volatility and makes it a sharper indicator of turning points and a closer companion to unemployment. We do not have an explanation for why the two series tracked each other so closely before 2008 and never again as closely afterward.

²⁵We seasonally-adjust the data as described in Section 3 and compare with the seasonally-adjusted JOLTS, so there may be differences in the seasonal adjustment procedure.

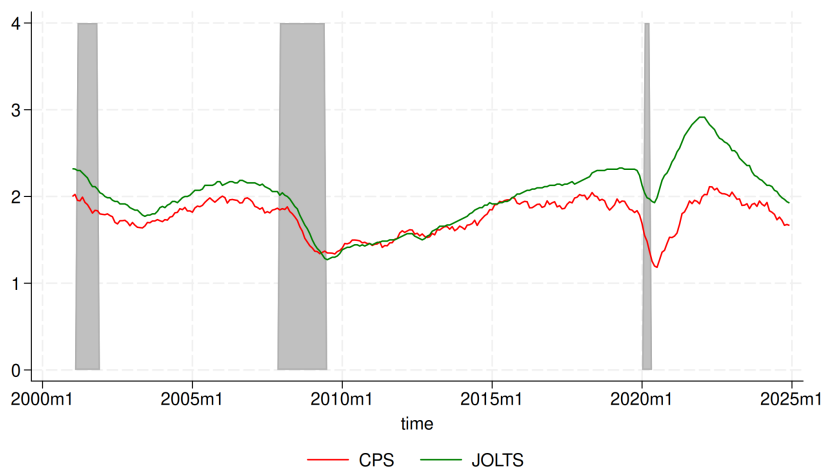


Figure 11: **Quits: CPS versus JOLTS**

Figure 11 makes the same comparison for quits, where the two series are more alike. Their levels are similar through about 2015: between 2001 and 2015 the average quit rate in our CPS data is 1.70, a little below the 1.82 rate in JOLTS. The series then diverge as labor markets tighten into the post-GFC recovery, a divergence that widens in the tight markets after 2021 before narrowing again as markets cool, and their correlation over the full period is 0.74, higher than for layoffs. The divergence in tight markets is what the concepts predict: our data capture only quits into non-employment, whereas JOLTS captures all quits, including those directly to another job, and such employer-to-employer quits rise precisely when vacancies are plentiful. The post-pandemic gap is the sharpest case: JOLTS quits reached an all-time high, giving rise to the “Great Resignation” narrative, while our series shows an increase in line with previous expansions. We cannot attribute the entire gap to employer-to-employer quits with certainty; other potential contributors include quits by multiple-job-holders, which JOLTS counts and we do not, and mismeasurement arising from the sharp decline in survey response rates over this period, particularly in JOLTS.

Taken together, the comparison is best read as a decomposition of the same underlying churn along different cuts, not as two estimates of the same concept. For the reason-by-destination composition of separations into non-employment, and for real-time measurement that is essentially unaffected by revision, our series is the appropriate measure; for the total volume of churn including on-the-job moves, JOLTS remains the broader instrument. The differences between them are differences of concept, most importantly that JOLTS quits and layoffs each contain a direct employer-to-employer component that ours does not, and that this component is neither small nor exclusively quits.

8 Conclusion

This paper shows that standard measures of gross worker flows mischaracterize job separations. The conventional approach treats flows from employment to unemployment as layoffs and flows to non-participation as quits. This assumption is inaccurate in both directions. Many laid-off workers exit the labor force rather than search, and many separations to non-participation are involuntary. Correcting the measurement has substantial consequences. Layoffs are larger and more cyclical than the employment-to-unemployment flow implies, and they account for more of the cyclical variation in both unemployment and employment. The negative correlation between quits and layoffs provides a stabilizing force that coarser flows obscure. The countercyclical labor-force attachment of laid-off workers, in turn, amplifies the volatility of unemployment. These findings lead us to recommend that researchers study labor-market churn and slack using quit and layoff series disaggregated by destination, rather than the standard flows into and out of unemployment.

These data open several avenues for future research. One is longer-run trends in job stability and mobility, such as the secular decline in quits and layoffs relative to tightness that we document. Another is heterogeneity: disaggregating by demographics, industry, occupation, or tenure would show which workers drive aggregate patterns, informing structural theories of firm-worker matching, specific human capital, and technological change. A third critical avenue is policy design, as knowing which workers exit the labor force after a layoff—and how this varies over the cycle—is directly relevant to the target calibration of unemployment insurance and retraining programs. Crucially, the real-time structural properties of our data series offer a distinct practical advantage for research over existing benchmarks. Unlike the JOLTS—which is constrained by an inability to observe the destination of separations, requires complex alignment methods, and is heavily revised—our series is available monthly, updated in near real-time, and remains unrevised. It similarly avoids the multi-year publication lags and retrospective friction inherent in smaller longitudinal panels like the PSID. Because this series extends back to 1978, it provides a stable, five-decade baseline spanning six recessions, allowing researchers to evaluate current labor market volatility in real time without sacrificing compositional accuracy or structural integrity.

During the preparation of this work the author(s) used Claude in order to aid the formatting of graphs. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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A Glossary

This section gives an overview of the different transition rates and the commonly used abbreviations.

For flow rates that follow the pattern origin-destination, each rate is abbreviated with two letters where the first one indicates the origin and the second indicates the destination:

- Employment-to-Unemployment (EU)
- Employment-to-Nonparticipation (EN)
- Unemployment-to-Employment (UE)
- Unemployment-to-Nonparticipation (UN)
- Nonparticipation-to-Employment (NE)
- Nonparticipation-to-Unemployment (NU)

For the transition rates which in addition to destination also specify the reason, each rate is abbreviated with three letters in the order origin, destination, reason:

- Employment-to-Unemployment due to a Layoff (EUL)
- Employment-to-Unemployment due to a Quit (EUQ)
- Employment-to-Nonparticipation due to a Layoff (ENL)
- Employment-to-Nonparticipation due to a Quit (ENQ)

B Data Robustness

B.1 Interchangeability of rotation groups MIS 4 and MIS 8

The reason-for-separation question is administered only in month-in-sample (MIS) 4 and 8, the outgoing rotation groups. Because our analysis pools these two rotations, we test whether they are interchangeable along two margins. All tests use the matched EN leaver sample (prime age), run separately for the pre-1994 and post-1994 regimes because the reason question's universe changed at the 1994 redesign, and report both unweighted and `wtfinl`-weighted estimates.

Outcome (reported reason). Among leavers with a coded quit or layoff, we compare the layoff share across MIS 4 and MIS 8 (a linear-probability mean difference; a logit of the layoff indicator on a MIS-8 dummy yields identical inference).

Composition (observables). We compare the two rotations on age, sex, education (four categories), and major occupation and industry groups, via mean-difference tests for the continuous/binary variables, a χ^2 test for the education distribution, and a joint Wald test of all covariates.

Table 20 reports the results. Every difference is small and, with a single exception, statistically insignificant; The lone sub-0.05 statistic—the post-1994 education χ^2 ($p = 0.045$)—does not survive the joint test ($p = 0.515$ unweighted) and reflects the very large sample rather than an economically meaningful gap.

Table 20: Rotation groups MIS 4 vs MIS 8: outcome and composition tests

Difference (MIS 8 – MIS 4)	Pre-1994			Post-1994		
	Est.	SE	p	Est.	SE	p
<i>Panel A. Reported reason: layoff share (pp)</i>						
Unweighted	1.06	0.78	0.170	−0.15	0.72	0.838
Weighted	0.56	0.88	0.527	0.10	0.84	0.906
<i>Panel B. Observable characteristics</i>						
Age (years), unwtld.	−0.114	0.109	0.297	0.029	0.081	0.724
Age (years), wtd.	−0.081	0.124	0.513	0.053	0.094	0.577
Female share (pp), unwtld.	−0.51	0.51	0.315	0.32	0.43	0.459
Female share (pp), wtd.	−0.89	0.60	0.140	0.00	0.51	0.993
<i>Panel C. Joint and distribution tests</i>						
Education, $\chi^2(3)$	0.52	—	0.914	8.05	—	0.045
All covariates, $F(22, \cdot)$, unwtld.	0.94	—	0.535	0.96	—	0.515
All covariates, $F(22, \cdot)$, wtd.	1.24	—	0.205	1.27	—	0.175

Notes. Matched EN leavers, prime age. Differences are MIS 8 minus MIS 4. Panel A is estimated on coded quit-or-layoff leavers; Panels B and C on all outgoing-rotation leavers. Weighted estimates use `wtfinl` sampling weights with robust standard errors. The joint F tests age, sex, education, and major occupation and industry groups simultaneously. The education χ^2 and joint F are omnibus tests with no single point estimate.

B.2 Representativeness of the outgoing rotation groups

Because the reason-for-separation question is asked only in the outgoing rotation groups (MIS 4 and 8), our series impute the ORG quit/layoff split onto all EN separations, including the four rotations never asked the question (MIS 2, 3, 6, 7). We assess whether this extrapolation is justified by comparing the two groups' observable characteristics. All comparisons use the matched EN leaver sample (prime age), are run separately for the pre- and post-1994 regimes, and report both unweighted and `wtfinl`-weighted estimates. We test age and sex via mean differences, the education distribution via a χ^2 test, and all characteristics jointly via a Wald test (occupation and industry entered as major groups).

Table 21 reports the results. The differences are uniformly small: ORG leavers are younger by at most about a quarter of a year, the female-share gap is below one percentage point, and the education distribution is statistically indistinguishable in both eras. Extrapolating the ORG

reason split to all rotations therefore introduces negligible composition bias.

Table 21: ORG (MIS 4/8) vs non-ORG (MIS 2/3/6/7) leavers: composition tests

Difference (ORG – non-ORG)	Pre-1994			Post-1994		
	Est.	SE	p	Est.	SE	p
<i>Panel A. Observable characteristics</i>						
Age (years), unwtd.	−0.126	0.064	0.050	−0.268	0.049	<0.001
Age (years), wtd.	−0.122	0.073	0.095	−0.282	0.057	<0.001
Female share (pp), unwtd.	0.02	0.30	0.957	−0.65	0.26	0.013
Female share (pp), wtd.	0.16	0.36	0.651	−0.81	0.31	0.008
<i>Panel B. Joint and distribution tests</i>						
Education, $\chi^2(3)$	2.06	—	0.561	4.05	—	0.256
All covariates, $F(22, \cdot)$, unwtd.	3.69	—	<0.001	3.05	—	<0.001
All covariates, $F(22, \cdot)$, wtd.	2.50	—	<0.001	2.94	—	<0.001

Notes. Matched EN leavers (prime-age). Differences are ORG minus non-ORG, split between pre-1994 and post-1994. Weighted estimates use `wtf1nl` sampling weights with robust standard errors. The joint F tests age, sex, education, and major occupation and industry groups simultaneously. The education χ^2 and joint F are omnibus tests with no single point estimate.

B.3 Non-response among ORG separations

This appendix supports Section 3.3. We first decompose the uncoded EN leavers into their components, describe how we separate record errors from genuine separations, and then characterize how the genuine uncoded separators differ from coded ones.

Decomposition Table 22 allocates all ORG EN leavers to coded reasons and to the components of non-response. Coded quit, layoff, and other shares are exact, as are the total no-code shares; the retired and disabled sub-shares are exact post-1994. The split of the remaining pool into record errors and genuine uncoded separations is estimated from respondents’ prior labor-force history (Appendix B.3), and is reported as a range.

Separating record errors from genuine separations The retired and disabled aside, the uncoded pool mixes two kinds of records that no observed variable distinguishes case by case: genuine EN separations whose reason went unrecorded, and spurious “employment” records that do not correspond to a real separation. We separate them using respondents’ pre-separation labor-force history in the matched panel, on the following logic. A genuine separator was employed before leaving, so in the preceding months looks like a coded leaver. A spurious record instead reflects a person who was already out of the labor force and was misclassified as employed; such records therefore carry excess nonparticipation in the months before the recorded separation.

Let π_g denote the share of group g whose history shows nonparticipation in the months preceding month t , for $g \in \{\text{coded}, \text{pool}\}$. Assuming genuine separators share the coded leavers’

Table 22: Decomposition of ORG $E \rightarrow N$ leavers (percent)

Category	Pre-1994	Post-1994	All years
Quit	36.3	22.5	27.3
Layoff	22.5	13.7	16.8
Other	18.2	13.1	14.9
<i>No reason code:</i>			
Retired (screened out by design)	—	4.7	3.1
Disabled (not asked)	—	7.3	4.8
Record error (spurious employment)	4.1–5.6	5.1–7.7	4.8–7.0
Genuine separation, reason not recorded	17.4–18.9	31.0–33.6	26.3–28.5
Total	100	100	100
N	26,755	50,005	76,760

Notes. Percent of all ORG (MIS 4/8) EN leavers (prime age). The retired and disabled screens do not exist pre-1994, so those rows are structurally absent, not merely unmeasured; the “all years” entries for them reflect post-1994 leavers only. Ranges on the last two rows are the bounds from the matched-history estimator (Appendix B.3); within a column they are jointly consistent with the total.

history distribution, and attributing the entire excess to spurious records, the spurious share of the pool is

$$\hat{\phi} = \pi_{\text{pool}} - \pi_{\text{coded}}.$$

We evaluate $\hat{\phi}$ under two definitions of “nonparticipation history”, which bracket the estimate: nonparticipant in *both* preceding months (N, N ; the tighter, lower bound) and in *either* month (any- N ; the upper bound). Because the procedure assigns *all* of the excess nonparticipation to record error, it maximizes the spurious share and hence treats the resulting genuine-separation share as a floor rather than a point estimate. Post-1994, the pool’s excess nonparticipation relative to coded leavers is 13.3 percentage points under the N, N definition and 19.8 under any- N , so record errors account for 13.3–19.8% of the pool—equivalently 5.1–7.7% of all ORG EN leavers, and the remainder, 31.0–33.6%, are genuine uncoded separations (Table 22). Pre-1994 there are no design screens, so the estimator is applied to the whole uncoded group: the excess is 17.9 (N, N) to 24.3 (any- N) percentage points, i.e. record errors are 17.9–24.3% of the uncoded, or 4.1–5.6% of leavers. Both calculations assume the sampled years’ history composition is representative of their era.

Composition of non-responders Table 23 compares coded leavers with the uncoded, differences oriented uncoded minus coded. Taken as a whole, non-responders differ sharply on every margin (Panel A): they are older, more likely female, and less educated. Much of this reflects the retired and disabled, however. Restricting to genuine uncoded separators—excluding the design-screened groups (Table 24, Panel A) and further holding prior employment history fixed (Panel B)—removes the age difference entirely post-1994, while the higher female share and lower education persist. Sex and education are thus the margins on which selection into reason-reporting operates, and the margins our reweighting targets.

Table 23: Coded vs uncoded ORG leavers: composition

Difference (uncoded – coded)	Pre-1994			Post-1994		
	Est.	SE	<i>p</i>	Est.	SE	<i>p</i>
<i>Panel A. Observable characteristics</i>						
Age (years), unwtd.	3.10	0.13	<0.001	1.61	0.08	<0.001
Age (years), wtd.	2.85	0.15	<0.001	1.53	0.09	<0.001
Female share (pp), unwtd.	8.33	0.60	<0.001	7.22	0.43	<0.001
Female share (pp), wtd.	8.38	0.66	<0.001	7.30	0.51	<0.001
<i>Panel B. Joint and distribution tests</i>						
Education, $\chi^2(3)$	118.2	—	<0.001	184.3	—	<0.001
All covariates, $F(22, \cdot)$, unwtd.	48.9	—	<0.001	47.7	—	<0.001
All covariates, $F(22, \cdot)$, wtd.	33.9	—	<0.001	32.0	—	<0.001

Notes. Matched ORG EN leavers (prime age) uncoded includes all leavers without a recorded reason (retired and disabled included). Weighted estimates use `wtfinal` with robust standard errors.

Table 24: Coded leavers vs genuine uncoded separators: composition

Difference (uncoded – coded)	Pre-1994			Post-1994		
	Est.	SE	<i>p</i>	Est.	SE	<i>p</i>
<i>Panel A. Excluding retired and disabled</i>						
Age (years), unwtd.	2.97	0.27	<0.001	0.09	0.25	0.709
Age (years), wtd.	3.13	0.31	<0.001	0.08	0.28	0.776
Female share (pp), unwtd.	7.97	1.31	<0.001	12.78	1.29	<0.001
Female share (pp), wtd.	8.49	1.44	<0.001	12.77	1.48	<0.001
Education, $\chi^2(3)$	22.9	—	<0.001	37.8	—	<0.001
All covariates, $F(22, \cdot)$, unwtd.	12.0	—	<0.001	8.1	—	<0.001
All covariates, $F(22, \cdot)$, wtd.	9.6	—	<0.001	6.1	—	<0.001
<i>Panel B. Verified stable employment history (E, E at $t - 3, t - 2$)</i>						
Age (years), unwtd.	2.92	0.44	<0.001	0.17	0.37	0.646
Age (years), wtd.	2.79	0.48	<0.001	0.39	0.42	0.357
Female share (pp), unwtd.	5.82	2.22	0.009	10.04	1.94	<0.001
Female share (pp), wtd.	7.08	2.40	0.003	9.41	2.21	<0.001
Education, $\chi^2(3)$	5.2	—	0.156	22.9	—	<0.001
All covariates, $F(22, \cdot)$, unwtd.	4.0	—	<0.001	3.5	—	<0.001
All covariates, $F(22, \cdot)$, wtd.	3.1	—	<0.001	2.8	—	<0.001

Notes. Sample of 1988–92 and 2014–18 separations (Apr–Dec), observables at $t - 1$. “Genuine uncoded separators” are leavers without a recorded reason, excluding those screened out as retired or disabled (Panel A); Panel B further restricts both groups to a verified employment history (E at $t - 3$ and $t - 2$), for whom the absence of a code cannot be a spurious record. Differences are uncoded minus coded; weighted estimates use `wtfinal` with robust standard errors. The joint F has 22 numerator d.f.; denominator d.f. range from 2,555 to 5,940 across subsamples. Group sizes (uncoded / coded): Panel A 1,425/4,538 pre, 2,592/2,863 post; Panel B 469/2,156 pre, 1,058/1,520 post.

B.4 Reweighted Results

The published series build the quit/layoff split from monthly unweighted counts of coded ORG leavers. Writing n_q, n_ℓ, n_o for the counts of coded quits, layoffs, and “other” in month t , the quit fraction is $qf_t = n_q/(n_q + n_\ell + n_o)$, and imputed quits are $qf_t \times \text{EN}_t$, where EN_t is the count of all EN leavers across the six rotations; layoffs are analogous. These flows are then seasonally adjusted and smoothed with a centered six-month moving average. Our correction changes only the fraction qf_t ; the imputation base EN_t and every downstream step are left exactly as in the baseline. Each coded leaver i receives a weight $w_i = 1/\hat{p}_i$, and the quit fraction becomes

$$qf_t = \frac{\sum_{i \in \text{quits}} w_i}{\sum_{i \in \text{coded}} w_i},$$

and analogously for layoffs and other. The propensity $\hat{p}_i$ is fitted by an era-specific logit (pre- and post-1994 estimated separately) on the full vector of observables measured at $t - 1$: age and its square, sex, education (four categories), and major occupation and industry groups. Weights are pure inverse propensities; consistent with the unweighted published construction, survey weights are not used. The specifications differ only in which propensity is inverted:

- **Baseline:** the published series, with no reweighting ($w_i = 1$).
- **M2 (rotation margin):** $\hat{p} = \text{Pr}(\text{ORG} \mid X)$, fitted on EN leavers in all six rotations, so that coded ORG leavers are reweighted to the observable composition of all rotations, i.e. correcting any ORG/non-ORG difference.
- **M1a (response margin, primary):** $\hat{p} = \text{Pr}(\text{reason recorded} \mid X)$, fitted on ORG leavers (excluding the post-1994 retired and disabled, screened out by design), correcting for selective non-response.
- **Both:** a single propensity $\text{Pr}(\text{ORG and reason recorded} \mid X)$ fitted over all EN leavers (excluding the post-1994 design-screened non-responders), combining the two corrections in one step.
- **Rolling M1a (36- and 12-month):** the M1a propensity re-estimated within moving windows that never cross the 1994 redesign, so that selection is allowed to vary over time.

Each table below reports the same statistics under these specifications.

Table 25: Average separation rates and quit–layoff correlation, by specification

Specification	Total	Quits	Layoffs	Corr(Q, L)
Baseline	2.12	0.85	1.27	−0.46
M2: rotation margin	2.12	0.85	1.27	−0.45
M1a: response margin	2.12	0.85	1.27	−0.49
Both (M1a + M2)	2.12	0.85	1.27	−0.48
Rolling M1a, 36-month	2.12	0.85	1.27	−0.48
Rolling M1a, 12-month	2.12	0.85	1.27	−0.48

Notes. Prime-age (25–55) monthly rates as a percent of employment, 1978m1–2024m12; the quit–layoff correlation is computed on the 6-month moving-average series, matching the published construction. Specifications are defined above.

Table 26: Destination shares and share correlation, by specification

Specification	Quits		Layoffs		Corr(s_{QN}, s_{LN})
	to U	to N	to U	to N	
Baseline	15.2	84.8	65.1	34.8	0.38
M2: rotation margin	15.2	84.8	65.0	34.9	0.36
M1a: response margin (primary)	15.1	84.9	65.1	34.8	0.34
Both (M1a + M2)	15.1	84.9	65.1	34.8	0.34
Rolling M1a, 36-month	15.1	84.9	65.1	34.8	0.32
Rolling M1a, 12-month	15.1	84.9	65.1	34.8	0.32

Notes. Percent of each flow reaching unemployment (U) versus nonparticipation (N); s_{QN}, s_{LN} are the quit- and layoff-to- N share series, correlated on their 6-month moving averages. The to- U split is `whyunemp`-based and is unaffected by the reweighting by construction. Specifications are defined above.

Table 27: Correlation of each flow with the unemployment rate, by specification

Specification	Quits			Layoffs			Tot. sep.
	to U	to N	Total	to U	to N	Total	
Baseline	−0.16	−0.48	−0.46	0.82	0.75	0.84	0.72
M2: rotation margin	−0.16	−0.48	−0.46	0.82	0.75	0.84	0.72
M1a: response margin (primary)	−0.16	−0.52	−0.50	0.82	0.75	0.84	0.72
Both (M1a + M2)	−0.16	−0.52	−0.50	0.82	0.75	0.84	0.72
Rolling M1a, 36-month	−0.16	−0.52	−0.50	0.82	0.76	0.84	0.72
Rolling M1a, 12-month	−0.16	−0.52	−0.51	0.82	0.76	0.84	0.72

Notes. Contemporaneous correlations of each seasonally adjusted, 6-month moving-average flow rate with the prime-age unemployment rate. Only the quit-to- N (and hence total-quit) flow moves under the correction; the layoff and to- U flows are unaffected by construction. Specifications are defined above.

B.5 DeNUNifying the Data

Table 28 shows that recoding potentially misclassified transitions has no effect on the main statistics in this paper.

Statistic	Main Data	DeNUNified Data
Averages		
Quits	0.84	0.84
Layoffs	1.27	1.27
Total Separations	2.11	2.11
Layoffs share N	0.40	0.40
Quit share N	0.85	0.85
EN	1.54	1.54
EU	1.08	1.08
Correlation with Unemployment Rate		
EUQ	-0.1478	-0.1464
ENQ	-0.3897	-0.3904
EQ	-0.3929	-0.3917
EUL	0.6013	0.6011
ENL	0.4885	0.4872
EL	0.6117	0.6119
EN	0.0122	0.0119
EU	0.6210	0.6213
Layoff share N	-0.6222	-0.6239
Quit share N	-0.1253	-0.1302
Corr(EQ,EL)	-0.3019	-0.3007

Table 28: Comparison with deNUNified data

B.6 Choice of moving average filter

In the main text, all timeseries were smoothed using a 6-month centered moving average smoother. We will show in the following that the choice of the smoothing parameter as well as whether it is centered or not does not change the data in any significant way. The following figures plot our layoff series, quit series, and total separations for four different smoothing techniques. 6-month centered is the standard we use in the main text, which means we include the previous 3 months, the current month, and three forward terms. 3-month, 4-month, and 6-month only include the previous 3, 4, and 6 months respectively, as well as the current month.

We see that the different lengths really only affects the pandemic period as it was so short but so extreme. It does not seem to affect other recessions or expansions. We checked including both lags and leads versus only including leads to make sure the most recent data is not significantly affected by the moving average filter. As we can see in the following figures, we see no difference between the two methods for the most recent observations.

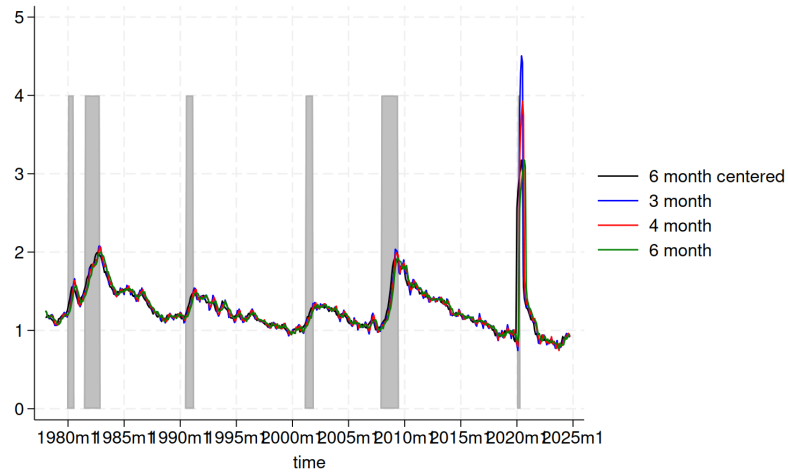


Figure 12: Comparison of moving-average filter for layoff series

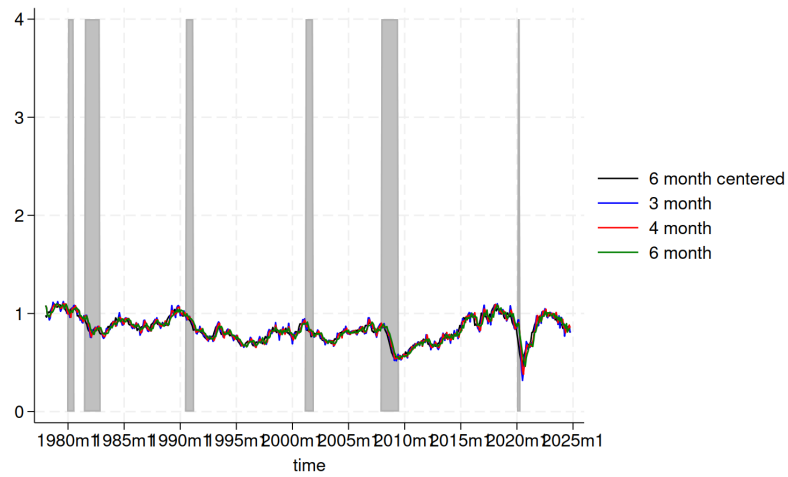


Figure 13: Comparison of moving-average filter for quit series

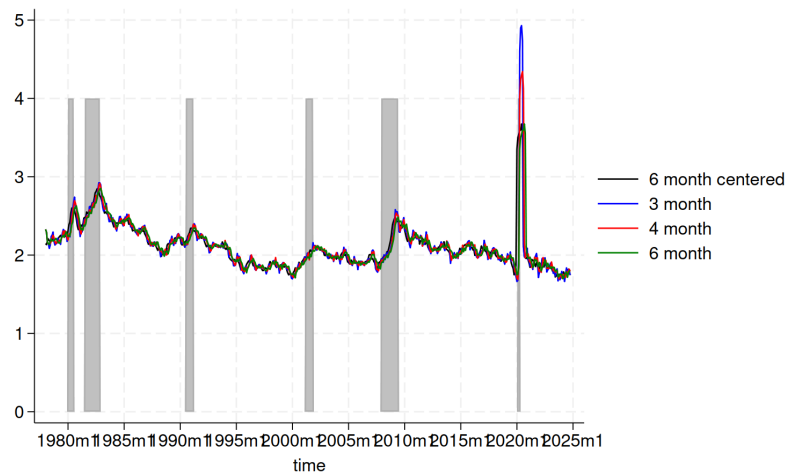


Figure 14: Comparison of moving-average filter for total separations

B.7 Permanent vs. Temporary Layoffs

When we harmonize the data to compute the layoff rate, the possible answer choices of unemployed and non-participating individuals in the CPS can be grouped in two categories: layoffs from a temporary job or from a permanent job. The former category includes everyone who reports losing their job because a temporary, seasonal, or intermittent job ended. The latter includes all other job losers.

Figure 15 shows that the business cycle pattern of layoffs is driven by permanent layoffs and the majority of layoffs are from permanent jobs. Interestingly, layoffs from temporary are mildly procyclical, i.e. decline during periods when unemployment is high. Because permanent layoffs

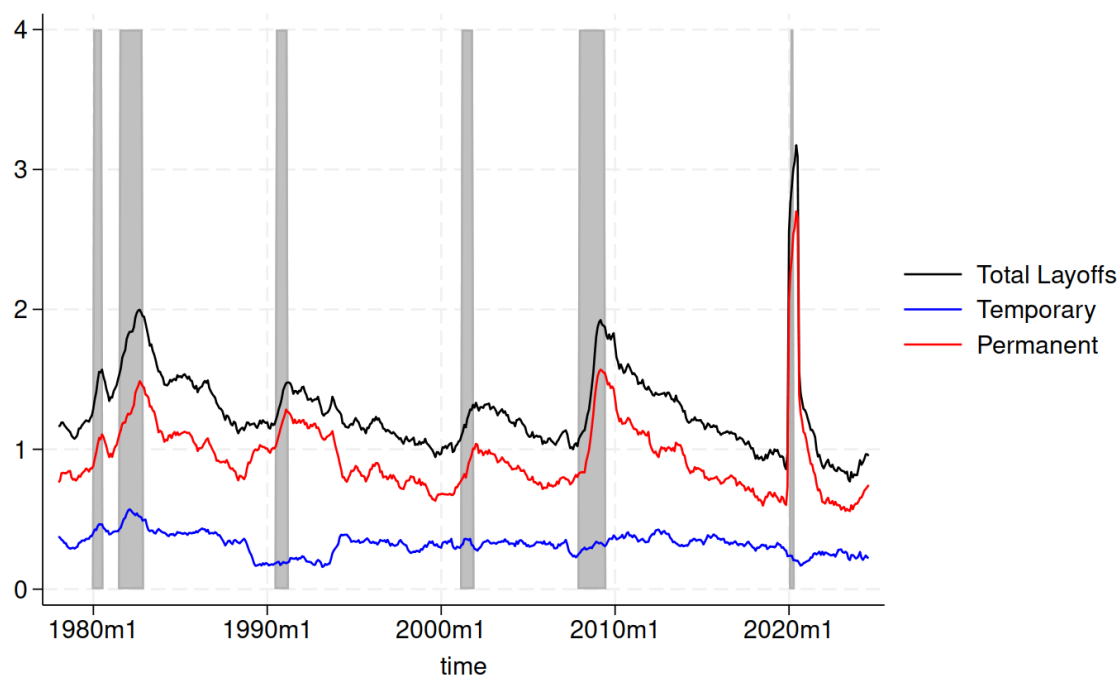


Figure 15: Layoffs from temporary vs permanent jobs

are strongly countercyclical and temporary layoffs mildly procyclical, recessionary periods are characterized by an increase in the share of layoffs from a permanent job.

C Statistics for Working Age Population.

In this section we replicate key statistics for the working age population. We continue to caution the reader that we cannot distinguish the reason for separation for people who flow from employment to non-participation and report that they are retired. These individuals are not asked why they left their prior job.

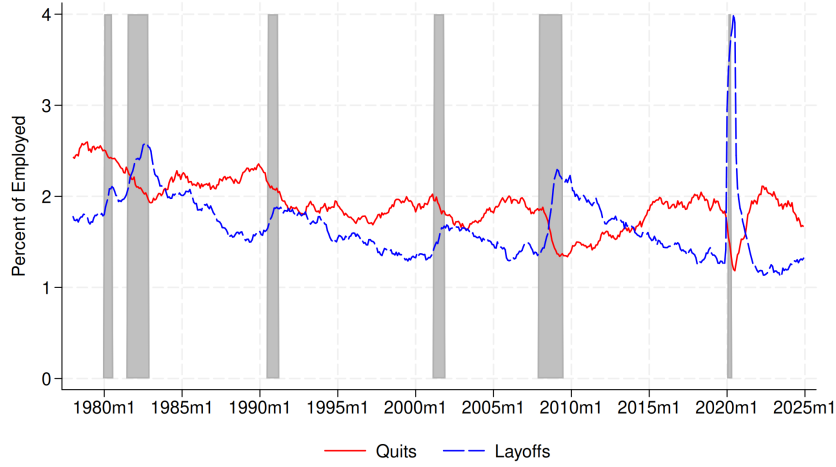


Figure 16: **Monthly Quits and Layoff rates to Non-Employment** (Shown as a percent of employment. Monthly seasonally-adjusted data and 6-month centered moving average)

	Total	Quits	Layoffs	Corr(Q,L)
1978-2024	3.66	1.92	1.74	-0.1222
1978-1990	4.16	2.24	1.92	-0.47
1991-2019,2021-2024	3.41	1.79	1.62	0.04

Table 29: Working Age: Monthly average percent of employed workers separating to non-employment by reason.

	Total	EN	EU	Corr(EN,EU)
1978-2024	4.45	3.11	1.35	0.31
1978-1990	5.10	3.48	1.63	0.10
1991-2019,2021-2024	4.21	2.96	1.24	0.31

Table 30: Working Age: Monthly average percent of employed workers separating to non-employment by destination.

	% Quits to		% Layoffs to		Corr(sQN,sLN)
	U	N	U	N	
1978-2024	9.1	90.9	55.6	44.4	0.14
1978-1990	10.8	89.2	55.0	45.0	-0.49
1991-2019,2021-2024	8.5	91.5	56.0	44.0	0.17

Table 31: Working Age: Monthly average share of quits and layoffs to non-employment by destination.

Recession	UR	EPR	Quits	Layoffs	Total Separations
1980	32.20%	-2.00%	4.23%	29.58%	18.00%
1981	50.00	-3.05	-15.71	26.66	10.56
1990	50.00	-2.22	-20.45	14.33	-1.53
2001	61.54	-3.26	-22.02	21.70	1.40
2007	112.77	-6.99	-37.17	74.14	22.77
2020	277.14	-13.48	-25.11	11.60	3.52

Table 32: **Comparing Recessions.** Percentage peak-to-trough change

Recession	Quits		Layoffs	
	Percentage points		Percentage change	
1980	-0.04	0.38	-4.32%	32.23%
1981	-0.19	0.67	-19.53%	49.39%
1990	-0.23	0.31	-23.28%	26.38%
2001	-0.16	0.36	-17.77%	36.83%
2007	-0.34	0.93	-38.51%	92.98%
2020	-0.42	2.13	-47.16%	242.99%

D Additional Statistics

D.1 Tight Labor Markets

We construct an indicator of tight labor markets to compare the dynamics of quits and layoffs in expansionary periods of the economy, i.e tight labor markets. We follow [Aaronson et al. \(2019\)](#) and use an indicator that equals one if the unemployment rate in a given quarter is below the noncyclical rate of unemployment and zero otherwise.²⁶

Statistic	Quits			Layoffs			Total sep.
	to U	to N	Total	to U	to N	Total	to (U+N)
$\text{Corr}(x, y)$	0.1828	0.2967	0.2664	-0.3491	-0.2291	-0.3880	-0.1945

Table 33: **Business Cycle Correlations.** Correlations of each flow (x) with the indicator for a tight labor market (y)

Table 33 shows the correlations of the different flow rates with the indicator of a tight labor market. They capture the movement of flows in exceptionally good times for comparison with Table 6 which uses the unemployment rate as an indicator. We see that tight labor markets (expansions) are characterized by higher quits, both into unemployment and non-participation. In other words, workers become more likely to quit when vacancies are high, the unemployment rate is low, and jobs are easy to find. Layoffs show the opposite pattern. Layoffs decline in tight

²⁶Specifically, we use the Noncyclical Rate of Unemployment released by the U.S. Congressional Budget Office, and retrieved from FRED (NROU). These data are at the quarterly frequency and so we transform our monthly data into quarterly by averaging over each quarter.

labor markets, which is not surprising.

Total separations, quits plus layoffs, decline during tight labor markets. Although quits increase, the fall in layoffs during these periods dominates the rise in quits and leads to an overall fall in separations into non-employment.

E Panel Study of Income Dynamics (PSID)

Our sample is PSID heads and spouses/partners in the 2003-2021 surveys. Similar to the CPS, we construct a panel with both the reason a job ended together with the reported year and month of the separation. These items are collected separately for the household head and for the spouse/partner; and from 2003 onward, for up to four jobs held rather than for the main job alone.

The monthly file is restricted to the waves from 2003-2021. This restriction is due to data availability. The month, year, and reason of separation are recorded for all four reported jobs only beginning in 2003; in earlier waves these fields exist for the single main job alone (and the month field for that main job is itself unavailable before 1985). Because a given wave captures job endings occurring over the roughly two preceding years, the resulting sample of dated separations spans approximately 2001–2019, with the COVID year of 2020 omitted by choice.

The classification of reason for separation uses the question on why a job ended (e.g. variable ER25431 for the spouse in the 2005 wave). The question is interviewer-coded from an open response prompted as follows (spouse wording, 2005 wave):

DE51. Why did she stop working for (NAME OF EMPLOYER)? Did the company go out of business, was she laid off, did she quit, or what?

The head version is identical in structure. Responses are coded into the categories reproduced in Table 34.

Table 34: Response codes for the “why last job ended” item

Code	Text	Our classification
1	Company folded / changed hands / moved; employer died / out of business	Layoff
2	Strike; lockout	Dropped
3	Laid off; fired	Layoff
4	Quit; resigned; retired; pregnant; needed more money; wanted a change	Quit
7	Other; transfer; any mention of armed services	Dropped
8	Job was completed; seasonal work; was a temporary job	Layoff
9	DK; NA; refused	Dropped
0	Inap. (no such person / did not work / still with employer, etc.)	Dropped

Placing each separation on a monthly timeline requires the month in which the job ended (variables `hhMoEnd`/`hwMoEnd`), together with the end year. A monthly separation date is formed

as the year–month combination for every observation classified as a quit, layoff, or temporary-job ending, and duplicate person-months are removed so that each individual contributes at most one separation per calendar month.

We merge the dated separations onto a monthly employment panel (providing monthly employment, unemployment, and out-of-labor-force status and the number of jobs held). Using the labor-market state in the separation month and the following month, each quit and each layoff is assigned to one of three mutually exclusive destinations: out of the labor force (N), unemployment (U), or employment (E , i.e. a job-to-job transition, restricted to workers holding a single job so as to capture a genuine employer change). This yields the six categories QN , QU , QE for quits and LN , LU , LE for layoffs. All tabulations use PSID longitudinal sampling weights.

Supplementary tabulations. The main text reports destination shares for prime-age workers. Table 35 reports the corresponding shares for all employed workers, over the full sample and over the Great Recession (2008–2010); the two share columns within each period differ only in their denominator, expressing each category either as a fraction of all quits (layoffs) or as a fraction of quits (layoffs) ending in non-employment. Table 36 reports the unweighted cell counts underlying the prime-age tabulations of the main text, again for the full sample and the recession. The shares in Table 37 are survey-weighted means of destination indicators within each separation type, with standard errors clustered on the individual to reflect repeated spells for the same worker.

Table 35: Destination shares of quits and layoffs, all employed workers

	Full sample		Great Recession	
	Share of all sep.	Share to nonemp.	Share of all sep.	Share to nonemp.
<i>Panel A: Quits</i>				
Out of labor force (QN)	0.564	0.912	0.660	0.914
Unemployed (QU)	0.054	0.088	0.062	0.086
Employed (QE)	0.398	—	0.294	—
<i>Panel B: Layoffs</i>				
Out of labor force (LN)	0.359	0.453	0.286	0.343
Unemployed (LU)	0.434	0.547	0.550	0.657
Employed (LE)	0.208	—	0.164	—

Notes. All employed workers, PSID waves 2003–2019; Great Recession columns restrict to separations dated 2008–2010. “Share of all sep.” expresses each destination as a fraction of all quits (Panel A) or all layoffs (Panel B): e.g. $QN/(QN+QU+QE)$. “Share to nonemp.” expresses it as a fraction of quits (layoffs) ending in non-employment: e.g. $QN/(QN+QU)$; it is undefined for the employment destination.

Table 36: Cell counts underlying prime-age tabulations (ages 25–54)

	Full sample	Great Recession
<i>Panel A: Quits</i>		
Out of labor force (QN)	2,968	571
Unemployed (QU)	423	64
Employed (QE)	3,271	357
<i>Panel B: Layoffs</i>		
Out of labor force (LN)	1,602	352
Unemployed (LU)	2,921	894
Employed (LE)	1,297	246

Notes. Unweighted counts of prime-age (25–54) separations by destination, PSID waves 2003–2019. The Great Recession column restricts to separations dated 2008–2010.

Table 37: Destination shares of quits and layoffs, prime-age workers

	Full sample		Great Recession	
	Share of all sep.	Share to nonemp.	Share of all sep.	Share to nonemp.
<i>Panel A: Quits</i>				
Out of labor force (QN)	0.467 (0.010)	0.878 (0.008)	0.548 (0.024)	0.877 (0.020)
Unemployed (QU)	0.065 (0.004)	0.122 (0.008)	0.077 (0.013)	0.123 (0.020)
Employed (QE)	0.488 (0.010)	—	0.394 (0.023)	—
<i>Panel B: Layoffs</i>				
Out of labor force (LN)	0.290 (0.011)	0.377 (0.013)	0.232 (0.018)	0.284 (0.021)
Unemployed (LU)	0.480 (0.012)	0.623 (0.013)	0.586 (0.021)	0.716 (0.021)
Employed (LE)	0.230 (0.010)	—	0.182 (0.017)	—

Notes. Prime-age workers (25–54); Great Recession columns restrict to separations dated 2008–2010. Standard errors, clustered on the individual in parentheses. Share definitions as in Table 35.

F Participation Neutral Series- Robustness

	Actual	Participation Neutral	Actual 6mo MA	Participation Neutral 6mo MA
Std. deviation (monthly)	0.0121	0.0102	0.0120	0.0101
Corr. with GDP cycle (quarterly)	-0.2912	-0.2353	-0.2974	-0.2401
Corr(Participation Neutral, Actual)		0.9785		0.9848

GDP cycle: HP filter ($\lambda = 1600$). 6mo MA: 6-month centered moving average.

Table 38: Business-cycle statistics: actual unemployment vs. participation neutral, 1978–2025 (excl. 2020).